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Liquidity Shocks, Business Cycles and Asset Prices ^{*}

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Abstract

In the aftermath of the Great Recession, macro models that feature financing constraints have attracted increasing attention. Among these, [Kiyotaki and Moore \(2012\)](#) is a prominent example. In this paper, we investigate whether the liquidity shocks and financial frictions proposed by [Kiyotaki and Moore \(2012\)](#) can improve the asset pricing predictions of the frictionless RBC model. We study the quantitative business cycle and asset pricing properties in an economy in which agents feature recursive preferences, are subject to a liquidity constraint, and suffer liquidity shocks. We find that the model predicts highly nonlinear time variation and levels of risk premia, which are driven by endogenous fluctuations in equity prices. However, the model fails to account for a basic fact: Periods of scarce liquidity are associated with high asset prices and low expected returns.

JEL classification: E22, E32, E44, G12

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1 Introduction

In the aftermath of the Great Recession, there has been increased interest by academics and policy makers in macro models that feature financing constraints. The Kiyotaki and Moore (2012) model (hereafter KM) of collateral constraints and liquidity shocks is a leading example. The main idea is that liquidity shocks constrain the fraction of assets that may be traded in a given period. These changes in liquidity can lead to fluctuations in aggregate activity and asset prices by tightening firms' ability to pledge collateral. Numerous studies have followed KM's lead (see Ajello (2016), Bigio (2015), Del Negro et al. (2016), Kurlat (2013), Venkateswaran and Wright (2014)).

In this paper, we study the quantitative properties of liquidity shocks in a real business cycle (RBC) framework, focusing on asset pricing properties and business cycle implications. To accomplish this, we use a stripped-down version of KM, characterize the equilibrium, and study the effects of liquidity shocks via global nonlinear analysis.

Our main findings are that liquidity shocks: (i) improve quantitative prediction of the levels and volatility of equity premiums relative to the frictionless RBC; (ii) predict highly nonlinear dynamics for premiums akin to models that feature balance sheet dynamics; (iii) have negligible effects on the risk-free rate; (iv) improve the relative volatility of investment to output growth rate; (v) on impact, have mild effects on the levels of investment and output; and (vi) produce counterfactual dynamics for the correlation between liquidity and the equity premium—i.e., periods of abundant liquidity are associated with higher expected returns. Lastly, after we decompose expected returns into a liquidity component and a market component, we find that the liquidity component does not account for a large share of the total premium. We demonstrate, in detail, why liquidity shocks fail to account for the basic fact that in tranquil times—which are typically associated with abundance in liquidity—expected excess returns are high.

The main mechanism in the KM framework is as follows. Investment has two characteristics that cause liquidity to become a source of business cycles. First, access to investment projects is limited to a fraction of the population, which means that resources must be reallocated from agents who do not possess these opportunities to those who do. Such reallocation requires a credible promise to deliver investment projects, which in turn requires collateral. Without collateral (or, in the model, less liquidity), that process is interrupted. Second, repayment cannot be guaranteed. This characteristic requires that investment be financed, in part, internally (investment requires a down payment). The combination of these two features creates gains from trading existing assets, and doing so enables agents to obtain internal funds to relax external financing constraints. Liquidity shocks interrupt the amount of trade, which affects the supply side of credit. In other words, periods of low liquidity are associated with a contraction in the supply schedule of claims to investment projects. When liquidity shocks are sufficiently large, they drive aggregate investment below its frictionless level. As a result, these shocks also drive a wedge between the price of capital and its replacement cost, which is a measure of inefficiently low investment.

From an asset pricing perspective, we highlight the fact that the model predicts highly nonlinear dynamics for risk premiums, together with sensible endogenous time variation. It also predicts higher levels

of premiums relative to those implied by the frictionless benchmark. In other words, the expected change in the price of equity is very sensitive to liquidity shocks. These elements of the model can significantly improve quantitative asset pricing predictions relative to the frictionless RBC setup.

The KM framework faces some challenges that we underscore throughout the paper. From a business cycle perspective, liquidity shocks produce countercyclical consumption—it increases in the recession triggered by a liquidity shock. However, the quantitative effect of this shock on macro variables is small overall: On impact, a major liquidity dry-out produces a decrease in output of -0.1% from its mean. Regarding asset prices and returns, the model predicts lower expected returns during periods of scarce liquidity. Thus, the model is counterfactual in the sense that tranquil times (those of abundant liquidity) are associated with high expected returns. Our results complement and reinforce other findings in the KM literature (i.e., Shi (2015)), but our focus on asset pricing offers further insight into the KM's implications for risk premiums dynamics.

We next provide a brief review of the literature.

Literature. Our paper is directly related to the literature that follows the KM model. Del Negro et al. (2016) is an example of this strand. A central element of that paper is the introduction of nominal rigidities and a monetary policy that is subject to a zero lower bound. Our paper complements theirs in the sense that we observe that without additional features, liquidity shocks cannot account for the dynamics of premiums and macro quantities. More precisely, we observe that liquidity shocks can improve frictionless predictions and come closer to the observed evidence, although this is not enough. In addition, our focus on risk premium dynamics emphasizes a central counterfactual element embedded in the KM constraint: In periods of abundant liquidity, expected excess returns are high. Lastly, our paper differs from Del Negro et al.'s because we study the behavior of the model globally, whereas theirs is restricted to a log-linearized version of the model.

Shi (2015) proposed an similar to ours, with a key difference in the analysis. In our paper, the labor supply schedule is fixed; liquidity shocks reduce capital accumulation, and this contracts labor demand. Thus, wages and hours decline in recessions. In contrast, Shi (2015) uses a “single family” framework with strong leisure-consumption complementarities. As in our setup, liquidity shocks lead to declines in the investment-consumption ratio. With leisure-consumption complementarities, labor supply contracts when liquidity shocks increase consumption. As a result, although liquidity shocks trigger strong reductions in hours, they also produce counterfactual movements in wages. In addition, our main focus is on asset prices and risk premium dynamics. We use recursive preferences Epstein and Zin (1989), and we calibrate both the elasticity of intertemporal substitution and the risk-aversion coefficient, as in the leading papers in the asset pricing literature. As noted by Shi (2015), our results also suggest a relatively higher unconditional level of the equity premium, but we extend the analysis to study its behavior across the state space.

Recent literature in macroeconomics has also predicted highly non-linear risk premium dynamics, driven by fluctuations in agents' balance sheets in models with financial frictions (Brunnermeier and San-nikov (2014) and He and Krishnamurthy (2013) are prominent examples). Our results are in this vein, but

with the crucial distinction that in our framework, endogenous fluctuations in expected returns are driven purely by occasionally binding constraints. This is because, as in KM, we assume that investment opportunities are i.i.d., and therefore we do not have to keep track of investor's balance sheets in the state vector.

The paper contains four sections. In Section 2, we present the model, characterize the solution, and discuss the intuition behind the effects. We explore the quantitative predictions of the model in Section 3, and Section 4 concludes.

2 Model

We begin by describing a version of the KM model, abstracting from fiat money. We consider an infinite-horizon economy in which time is discrete and denoted by $t = 0, 1, \dots$. There are two populations, entrepreneurs and workers, each with unit measure. The former do not work, but invest in physical capital, while the latter do not invest but supply labor. In each period, entrepreneurs are randomly assigned to one of two types: investors or savers, labeled by superscripts i and s .

The economy is subject to two sources of aggregate uncertainty: productivity and liquidity shocks. We represent the productivity level in the economy by $A_t \in \mathbb{A}$ and liquidity by $\phi_t \in \Phi \subset [0, 1]$. The nature of liquidity shocks will affect the ability to sell equity and will soon become clear. Importantly, we assume that both A_t and ϕ_t follow a finite-space stationary Markov process with a stationary transition probability $\Pi : (\mathbb{A} \times \Phi) \times (\mathbb{A} \times \Phi) \rightarrow [0, 1]$.

While ϕ_t and A_t are exogenous aggregate variables, there is endogenous capital accumulation. We denote the level of capital in the economy by $K_t \in \mathbb{K}$; this completes the description of the state space. Hence, we define $s_t = \{A_t, K_t, \phi_t\} \in S_t \equiv (\mathbb{A} \times \mathbb{K} \times \Phi)$ as the vector that represents a point in the state space.

We next describe the production side of the economy and then the entrepreneur's problem.

2.1 Production

A representative firm produces final consumption goods. The firm hires labor from workers and capital from entrepreneurs. Production is carried out according to a Cobb-Douglas technology

$$y_t = A_t F(k_t, L_t) ,$$

where y_t is output and $F(k, L) = k^\alpha L^{1-\alpha}$. The profit maximization problem is standard, and the firm demands $\{L^d, k^d\}$ to solve

$$\pi_t = \max_{L^d, k^d} A_t (k_t^d)^\alpha (L_t^d)^{1-\alpha} - w_t L_t^d - r_t k_t^d , \quad (1)$$

where w_t is the real wage and r_t the net return on capital.

2.2 Entrepreneurs and investment

Preferences. Entrepreneurs feature recursive preferences with Kreps-Porteus aggregator, as in [Epstein and Zin \(1989\)](#):

$$V_{j,t} = \left[c_{j,t}^\rho + \beta E_t \left[V_{j,t+1}^{1-\gamma} \right]^{\frac{\rho}{1-\gamma}} \right]^{\frac{1}{\rho}}. \quad (2)$$

In this notation, we use $\rho = 1 - 1/\psi$, where ψ is the elasticity of intertemporal substitution (EIS) and γ the risk-aversion parameter. We represent the certainty equivalent as $\mu_t = E_t \left[V_{j,t+1}^{1-\gamma} \right]^{\frac{1}{1-\gamma}}$. Notice that both types of entrepreneurs $j = (\text{savers}, \text{investors})$ have the same preferences, and the only source of heterogeneity is their investment opportunities.

Investment opportunities. Investment opportunities arrive randomly, are distributed i.i.d across time and agents, and are independent of aggregate shocks.¹ In particular, an investment opportunity arrives with probability π . We label entrepreneurs who have an investment opportunity “investors”—whose mass is π —and those without such opportunity “savers”—whose mass is $1 - \pi$. Entrepreneurs face the following budget constraint:

$$c_t + i_t^d + q_t \Delta e_{t+1}^+ = r_t n_t + q_t \Delta e_{t+1}^-. \quad (3)$$

The right-hand side of (3) corresponds to the resources available to the entrepreneur. The first term is the return to equity holdings, where r_t is the return on equity and n_t is the amount of equity held by the entrepreneur. The second term on the right is the value of equity sales, $q_t \Delta e_{t+1}^-$, where q_t represents the price of equity. This term is the difference between the next period’s stock of equity e_{t+1}^- and the nondepreciated fraction of equity owned in the current period λe_t^- . The entrepreneur uses these funds to consume c_t , to finance down payments for investment projects, i_t^d and to purchase outside equity $q_t \Delta e_{t+1}^+$. Each unit of e_t^- entitles other entrepreneurs to rights to the revenues generated by the entrepreneur’s capital, and e_t^+ entitles the entrepreneur to revenues generated by other entrepreneurs. Then, the net equity for each entrepreneur and the law of motion for capital are given by

$$n_t = k_t + e_t^+ - e_t^-, \quad (4)$$

$$k_{t+1} = \lambda k_t + i_t, \quad (5)$$

where i_t is the aggregate investment we define below. The sole difference between investors and savers is that the latter are not allowed to invest directly by using internal funds—they do not have an investment opportunity. That is, they are constrained to $i_t^d = 0$. Then, the law of motion for outside equity and issued

¹This assumption facilitates aggregation, as we explain below.

equity is given by

$$\begin{aligned} e_{t+1}^+ &= \lambda e_t^+ + \Delta e_{t+1}^+, \\ e_{t+1}^- &= \lambda e_t^- + \Delta e_{t+1}^- + i_t^s, \end{aligned} \quad (6)$$

and we next describe i_t^s , which is the entitlements to others entrepreneurs (given by an investment contract).

Budget and financing constraints. When an investment opportunity is available, entrepreneurs choose a scale for an investment project, i_t . Projects increment their capital stock one for one with the size of the project. Each project is funded by a combination of internal funding, i_t^d , and external funds i_t^f . External funds are obtained by selling equity that entitles other entrepreneurs to the proceeds of the new project. Thus, the following accounting identity holds: $i_t = i_t^d + i_t^f$. The investment project i_t can be also be divided into final ownership i_t^i of the investor and the ownership of other entrepreneurs, i_t^s . That is, $i_t = i_t^s + i_t^i$.

Since we assume that equity is homogeneous and equity markets are competitive, investors can raise $q_t i_t^s$ funds. Thus, external financing is equal to $i_t^f = q_t i_t^s$. Notice that at the end of the period, the investing entrepreneur increases his equity in $i_t^i = i_t - i_t^s$, while he has contributed only $i_t - q_t i_t^s$.

In addition to the accounting principles stated above, investment is subject to moral hazard: Investors can divert funds from projects. By diverting funds, they are able to increment their equity up to a fraction $1 - \theta$ of the total investment. If they divert funds, the remainder of the project is lost. No enforcement or commitment technologies are available. Therefore, the incentive compatibility condition for external financing is equivalent to

$$(1 - \theta) i_t \leq i_t^i \Rightarrow i_t^s \leq \theta i_t, \quad (7)$$

which means that outside equity holder's stake in the project may not be higher than θ . It is worth emphasizing that the distinction between inside and outside equity renders the KM model a q -theory of investment. The wedge occurs as a combination of two things: First, only a fraction of agents have access to investment opportunities, which generates a demand for outside equity. Limited enforcement causes the supply of outside equity to be limited by the incentive compatibility constraints. The value of q must adjust to equate demand with supply and this price may differ from 1.

In addition to the incentive compatibility constraint implied by moral hazard (7), there is a constraint on the sales of equity created in previous periods. Resellability constraints impose a limit the equity that can be sold at every period. These constraints depend on the liquidity shock ϕ_t

$$\Delta e_{t+1}^- - \Delta e_{t+1}^+ \leq \lambda \phi_t n_t. \quad (8)$$

Kiyotaki and Moore (2012) motivates these constraints by adverse selection in the equity market. Bigio (2015) and Kurlat (2013) show that such a constraint can follow from adverse selection in the quality of assets. There are multiple alternative explanations for why liquidity may vary over the business cycle. What matters here is that liquidity shocks ϕ_t prevent equity markets from materializing gains from trade. Our

focus is to study how shocks to liquidity can improve the predictions of business-cycle and asset-pricing theory without prying on where these shocks come from.

The investor's objective is to maximize her proceeds from total investment i_t^i by choosing a positive amount of i_t^s . We solve that program subject to the accounting identities $i_t = q_t i_t^s + i_t^d = i_t^i + i_t^s$, and, importantly, the constraint $i_t^s \leq \theta i_t$. Once we substitute the accounting identities, the investment decision—given by i_t^d —is characterized by

$$\begin{aligned} \max_{\{i_t^s\}} & i_t^d + (q_t - 1) i_t^s, \\ \text{st. } i_t^s & \leq \theta (i_t^d + q_t i_t^s). \end{aligned} \quad (9)$$

Interpretation of this problem is clear. For every project, the investing entrepreneur increases her stock of equity $i_t^d + (q_t - 1) i_t^s$, which is the sum of the down payment plus the gains from selling equity corresponding to the new project, i_t^s . The constraint says that the amount of outside funding is limited by the incentive compatibility constraint. If $q_t \in (1, \frac{1}{\theta})$, the problem is maximized at the points at which the incentive compatibility constraint binds. Therefore, within this price range, for every unit of investment i_t , the investing entrepreneur finances the amount $(1 - q_t \theta)$ units of consumption and owns the fraction $(1 - \theta)$. This problem characterizes investors' resource constraint, which we state in the following proposition.

Proposition 1 [Investors' resource constraint] *When $q_t > 1$, we can re-write investors' budget constraint as follows*

$$c_{i,t} + q_t^R n_{t+1} = (r_t + \lambda q_t^i) n_t,$$

where $q_t^R = \frac{1-\theta q_t}{1-\theta}$ and $q_t^i = \phi_t q_t + (1 - \phi_t) q_t^R$. When $q_t = 1$, the budget constraint is identical to that of savers, and is given by

$$c_t + q_t n_{t+1} = (r_t + \lambda q_t) n_t.$$

Proof. See appendix.

2.3 Agents' problems

Workers. We assume that workers are hand to mouth, and therefore face a static problem: choosing how much to consume and work. Their preferences are

$$U_t(c_{w,t}, L_t) = c_{w,t} - \frac{\kappa}{1+\nu} L_t^{1+\nu}, \quad (10)$$

and the budget constraint is just $c_{w,t} = w_t L_t^s$, where L_t^s is the supply of labor chosen to maximize U_t .

Savers. For savers, the problem consists of maximizing the value function (2) by choosing the level of

consumption and equity

$$\begin{aligned} V_{s,t} &= \max_{c_{s,t}, n_{s,t+1}} \left[c_{s,t}^\rho + \beta E_t \left[V_{s,t+1}^{1-\gamma} \right]^{\frac{\rho}{1-\gamma}} \right]^{\frac{1}{\rho}} \\ &\text{s.t.} \\ c_{s,t} + q_t n_{s,t+1} &= (r_t + q_t \lambda) n_t \equiv \omega_{s,t}. \end{aligned} \quad (11)$$

Investors. Similarly, investors maximize their value function, but their investment opportunity set is different from that of savers because their budget constraint encodes the optimal investment decision through the price q_t^i .

$$\begin{aligned} V_{i,t} &= \max_{i_t, c_{i,t}, n_{i,t+1}} \left[c_{i,t}^\rho + \beta E_t \left[V_{i,t+1}^{1-\gamma} \right]^{\frac{\rho}{1-\gamma}} \right]^{\frac{1}{\rho}} \\ &\text{s.t.} \\ c_{i,t} + q_t^R n_{i,t+1} &= (r_t + q_t^i \lambda) n_t \equiv \omega_{i,t}. \end{aligned} \quad (12)$$

We next define the equilibrium and elaborate on the model's solution.

2.4 Equilibrium

In this section, we first define equilibrium and characterize the optimality conditions for each class of agents. We then provide a decomposition of the returns on wealth.

Definition 2 [Recursive competitive equilibrium] A recursive competitive equilibrium is a set of price functions $\{q, w, r\}$, real quantities c_j, n_j^i, i_j, L for $j = s, i$, a sequence of distribution of individual equity holdings Λ_t , and a transition (law of motion) for the aggregate state of the economy Ξ , such that: (1) given prices, optimal policies solve agents problems (i.e., those of firms, workers and entrepreneurs); (2) goods market clears; (3) labor market clears at price w ; (4) equity market clears at price q ; (5) aggregate capital evolves as $K' = \lambda K + I$; and (6) both Λ and Ξ are consistent with the policy functions that solve agents' problems.

Notice that in the above definition, the distribution of equity holdings, Λ_t , is a relevant state at an individual level but does not determine aggregate variables. In other words, the distribution of equity is not an object in the aggregate state space. Intuitively, this is because of the assumption that investment opportunities are i.i.d across investors.

Worker's policy functions. The maximization of (10) is standard and yields

$$L_t^s = \left[\frac{w_t}{\kappa} \right]^{\frac{1}{\nu}},$$

where L^s stands for labor supply. Next, we characterize firms' decisions and equilibrium prices (w_t, r_t) .

Firms' equilibrium and equilibrium prices. From firms' maximization problem (1), labor demand is given by

$$L_t^d = \left[\frac{(1-\alpha) A_t}{w_t} \right]^{\frac{1}{\alpha}} k_t ,$$

thus, using equilibrium $L_t^d = L_t^s$, the equilibrium real wage is

$$\kappa^{\frac{\alpha}{\alpha+\nu}} [(1-\alpha) A_t]^{\frac{\nu}{\alpha+\nu}} k_t^{\frac{\alpha\nu}{\alpha+\nu}} = w_t^* .$$

Lastly, define the equilibrium return on capital, using firms' first-order conditions:

$$r_t^* = \alpha A_t \left[\frac{(1-\alpha) A_t}{\kappa} \right]^{\frac{(1-\alpha)}{\alpha+\nu}} k_t^{\frac{(\alpha-1)\nu}{\nu+\alpha}} . \quad (13)$$

Next, after solving firms' and workers' optimal policies, we characterize the solution for entrepreneurs.

Entrepreneurs. We define the right-hand side of savers' and investors' budget constraints as $\omega_t^s \equiv (r_t + q_t \lambda) n_t^s$ and $\omega_t^i \equiv (r_t + q_t^i \lambda) n_t^i$. From (2), we obtain the intertemporal marginal rate of substitution given by

$$M_{t,t+1}^j = \beta \left(\frac{c_{j,t+1}}{c_{j,t}} \right)^{\rho-1} \left(\frac{V_{j,t+1}}{\mu_{j,t}} \right)^{1-\gamma-\rho} ,$$

and therefore we need to solve for the value function. We proceed with a guess and verify strategy. The next proposition characterizes optimal policies for investors and savers:

Proposition 3 [Policy functions] (i) *Savers' optimal decision rules are*

$$\frac{c_{s,t}}{\omega_{s,t}} = 1 - \xi_{s,t} , \quad (14)$$

$$q_t n_{s,t+1} = \xi_{s,t} \omega_{s,t} . \quad (15)$$

(ii) *Investors' optimal decision rules are, when $q_t > 1$*

$$\frac{c_{i,t}}{\omega_{i,t}} = 1 - \xi_{i,t} , \quad (16)$$

$$q_t^R n_{i,t+1} = \xi_{i,t} \omega_{i,t} , \quad (17)$$

$$i_{t+1} = \frac{n_{i,t+1} + (\phi_t - 1) \lambda n_t}{(1 - \theta)} , \quad (18)$$

where $\xi_{s,t}$ and $\xi_{i,t}$ are unknown functions of the aggregate state of the economy that must be solved using equilibrium conditions. When $q_t = 1$, savers' and investors' policies are identical.

Proof. See appendix.

Returns. Following KM, we define the returns on wealth according to the alternative entrepreneurs' in-

vestment opportunity set.

$$R_{t+1}^{ss} = \frac{r_{t+1} + q_{t+1}\lambda}{q_t}, R_{t+1}^{is} = \frac{r_{t+1} + q_{t+1}\lambda}{q_t^R}, R_{t+1}^{ii} = \frac{r_{t+1} + q_{t+1}^i\lambda}{q_t^R}; R_{t+1}^{si} = \frac{r_{t+1} + q_{t+1}^i\lambda}{q_t}.$$

Here, R_{t+1}^{ss} is the return on wealth obtained by a saver who continues as a saver in the next period. In this case, the saver obtains—after holding $1/q_t$ units of equity—the net return on capital r_{t+1} plus the undepreciated portion of equity—valued at the spot price q_{t+1} . Similarly, R_{t+1}^{si} represents the return for a saver who receives an investment opportunity in the next period; recall this occurs with probability π . In this case, however, the undepreciated equity is valued at q_{t+1}^i , since he will face an investment opportunity and thus a weighted price: A fraction ϕ is non-resellable—valued at q_t^R —and a fraction $1 - \phi$ is resaleable at the spot price q_t . Both R_{t+1}^{is} and R_{t+1}^{ii} follow the same logic.

From here, we can decompose the conditional return on wealth for investors and savers. In particular, we compute the conditional return on wealth faced by entrepreneurs and decompose it between a “market value” that corresponds to fluctuations in market prices and a “liquidity component” that arises from the frictions in the model. The idea is to express all of the returns as a spread relative to a benchmark market return, R_{t+1}^{ss} . The return R_{t+1}^{ss} is the appropriate benchmark, because it is the market valuation of the future flow of capital dividends.

Let R_{t+1}^i be the investor’s return on wealth conditional on his individual state. In the next proposition, we characterize a decomposition of this return into a “market component” and a “liquidity component”

Proposition 4 [Decomposition] *The investor’s return on wealth R_{t+1}^i can be decomposed as*

$$R_{t+1}^i = \frac{l_{t+1}}{q_t^R} + \eta_t R_{t+1}^{ss}, \quad (19)$$

where $l_{t+1} = -\pi \frac{\lambda(q_{t+1} - q_{t+1}^R)(1 - \phi_{t+1})}{q_t^R}$ and $\eta_t = \left(\frac{(1-\theta)q_t}{1-\theta q_t} \right) \left(\equiv \frac{q_t}{q_t^R} \right)$. Also, the return on wealth faced by a saver, R_{t+1}^s , can be decomposed as

$$R_{t+1}^s = \frac{l_{t+1}}{q_t} + R_{t+1}^{ss} \quad (20)$$

Proof. See appendix.

We define the intercept term $l_{t+1} \leq 0$ as the “liquidity component”, which shows up both in (19) and (20), and captures the following intuition: There is a probability π that the entrepreneur—regardless of his initial type—will become an investor in the next period. In such a case, a fraction $(1 - \phi)$ of the λ remaining equity becomes illiquid. Thus, if he becomes an investor, his internal valuation of this illiquid equity is q_{t+1}^R instead of the spot price q_{t+1} ; the entrepreneur could have replaced that unit at price q_{t+1}^R . Since he values that illiquid portion at price q_{t+1}^R instead of q_{t+1} , a fraction $(1 - \phi)$ loses $(q_{t+1} - q_{t+1}^R)$ with probability π relative to the benchmark return. In either formula, the term l_{t+1} is scaled by either $1/q_t^R$ or

$1/q_t$ depending on which was the initial cost of capital—for investors and savers, respectively. The term $\eta_t \geq 1$ in (19) is a multiplier on the market return over the investor's portfolio. This term simply captures the idea that for this entrepreneur, it costs him q_t^R to build the capital that costs q_t in the market. The greater the discrepancy, the higher $q_t > 1$, the greater this multiplier. As we explained earlier, the internal cost for the investor $q_t^R \leq 1 \leq q_t$ captures the idea that the investor is effectively exploiting an arbitrage.

This concludes our specification of the model environment, definition of the equilibrium concept, and characterization of the optimal policies. We now briefly discuss about the role of liquidity in the model, then we investigate the quantitative properties of the model.

2.5 Discussion

The optimal financing problem subject to the enforcement constraints is essentially static; hence, we can use a static analysis to understand the effects of liquidity shocks. We illustrate the model's main mechanism using this approach.

Whenever $q_t > 1$, external financing allows investing entrepreneurs to arbitrage. In such instances, the entrepreneur wants to finance investment projects as much as possible. We have already shown that when constraints are binding, the entrepreneur owns the fraction $(1 - q_t)$ of the investment, but finances only $(1 - \theta q_t)$. If he uses less external financing, he misses the opportunity to obtain more equity. To gain intuition, it is convenient to abstract from the consumption decision and assume that the investor entrepreneur uses $x_t \equiv q_t \phi_t \lambda n_{i,t}$ to finance the down payment. Thus, $i_t^d = x_t$.

The constraints impose a restriction on the amount of external equity that may be issued, $i_t^s \leq \theta i_t$. External financing is obtained by selling i_t^s equity at a price q_t , so the amount of external funds for the project is $i_t^f = q_t i_t^s$. Since $i_t = i_t^f + i_t^d$, external financing satisfies

$$i_t^f \leq q_t \theta (i_t^f + x_t). \quad (21)$$

Figure 1 illustrates the simple intuition behind the liquidity channel—i.e., how changes in the amount x_t of sales of equity corresponding to older projects affect investment by restricting the amount of external financing for new projects. Panel (a) in Figure 1 plots the right- and left-hand sides of restriction on outside funding given by inequality (8). Outside funding, i_t^f , is restricted to lie in the area where the affine function is above the 45-degree line. Since $q_t > 1$, the left panel shows that the liquidity constraint imposes a cap on the capacity to raise funds to finance the project. Panel (b) shows the effects of a decreases in x_t without considering any price effects. The fall in the down payment reduces the intercept of the function defined by the right-hand side of Figure 1. External funding, therefore, falls together with the whole scale of the project. Since investment falls, the price of q must rise such that the demand for saving instruments falls to match the fall in supply. The increase in the price of equity implies that the amount financed externally is higher. The effect on the price increases the slope and intercept, which partially counterbalances the original effect. This effect is captured in panel (c) of Figure 1.

3 Quantitative analysis

In this section, we explore the models quantitative predictions by solving it numerically using a global solution method. In the following seven subsections, we first describe the calibration, then perform several exercises to shed light on all of the models properties.

Calibration. We divide parameters into three categories and fix the time period to a quarter. Table 1 reports the list of parameters and values. We set the capital share to $\alpha = 0.36$ and the depreciation rate $1 - \lambda = 0.025$; these two numbers are well established in the RBC literature. The arrival of investment opportunities π is set to 0.012, a slightly greater number than that of Del Negro et al. (2016). Using $\pi = 0.009$ as in their paper, implies that the economy spends considerable time in the constrained region—i.e., the enforcement constraint almost always binds. Since they solve their model by log-linearizing around a steady state in which financial constraints bind, it is reasonable for them to use a small value for π . In contrast, we choose a π that is close to theirs, but with this value the economy also spends some time in the unconstrained equilibrium. In other words, we exploit the global solution method to study an economy in which constraints occasionally bind. The calibrated π allows us to match the observed fact the vast majority of firms do not exhibit investment rates greater than 20%.² Next, we calibrate θ following the extensive literature that analyzes financing constraints à-la KM. We set a value of 0.77—very similar to that of Del Negro et al. (2016), who use 0.792. This implies that the upper bound for the spot price of equity q_t is $1/\theta \approx 1.3$. Notice that this parametrization satisfies $0 \leq \theta \leq 1 - \pi = 0.988$.

In terms of preferences, we calibrate β to 0.99 to match an annual real interest rate of around 1.7% at an annual frequency. The EIS and the risk-aversion parameters are from the asset-pricing literature, primarily following Bansal and Yaron (2004). In particular, we calibrate $\gamma = 10 > \psi = 1.5 > 1$, which implies that agents feature a preference for early resolution for uncertainty. We explore the implications for the EIS below. Lastly, the Frisch elasticity ($1/\nu$) is set to 1, which is commonly used in the RBC literature and is in range of the macro estimates of Chetty et al. (2011), and the parameter κ is calibrated to obtain an average 30% of working hours.

Finally, in terms of aggregate shocks, we first calibrate $\rho_A = 0.95$ and $\sigma_A = 0.01$, again following the RBC literature and the behavior of US productivity quarterly time series. The support of liquidity shocks (i.e., the maximum and minimum values for ϕ) is set to $[\phi_L, \phi_H] = [0.05, 0.5]$, which yields an unconditional mean liquidity of 0.275. This is a smaller value than that of Del Negro et al. (2016), who calibrate a steady-state level for liquidity at 0.4. However, it is worth emphasizing that the concept of liquidity in their paper differs from KM: They introduce a positive supply of government bonds to their analysis, and they calibrate the steady-state level of liquidity to match the convenience yield in the data. In this paper, instead, we choose a lower unconditional mean of liquidity, since we do not introduce government bonds. The last parameter for the aggregate shocks is the correlation between aggregate productivity A and liquidity ϕ , which we set to 0.75. Bigio (2015) and Kurlat (2013) study the relationship between liquidity and

²Cooper et al. (1999) show that, using the US longitudinal survey (which covers US manufacturing industries, but accounts for only about 45% of aggregate investment), 87% of firms display investment rates lower than 23%. The calibrated $\pi = 0.012$ implies that 6% of firms have investment rates between 17% and 28%.

aggregate productivity when the former is determined by the solution to a problem of asymmetric information. Both papers provide theoretical results in which the relation between productivity and liquidity is positive. It is worth emphasizing that quantitative results are not sensibly affected by this correlation.

[Table 1 about here: Calibration]

We finish the calibration section with the following observation. We purposefully avoid calibrating the three key parameters of KM's framework (i.e. $\theta, \pi, E[\phi]$) to match asset-pricing or business-cycle dynamics. Instead, we adopt the practice of setting parameters to reasonable micro evidence and evaluate the macroeconomic performance. This allows us to do a comprehensive assessment of KM without departing from previous results in the literature.

Equilibrium through the state space. We start with a description of the behavior of policy functions and endogenous variables across the state space. All figures display the stock of capital in the horizontal axis. Upper panels display the variable of interest in the 20% percentile of the productivity level ("low productivity states"), while lower panels display the 80% percentile ("high productivity states"). Similarly, left and right panels display the 20% and 80% percentiles of the liquidity level, respectively.

Figure 3 illustrates the spot price of equity q , the internal cost of capital for investors q^R , and their liquidity-weighted price q^I . The figure shows the inverse relationship between the spot price of capital and the internal cost. The logic, as explained above, is that investors own a fraction $1 - \theta$ of final investment whereas they only put $1 - q\theta$ of the funds in that project. This means that as q increases, they fund a smaller share of their projects, because they can sell claims to investment at a higher price. The figure then explains how q varies with TFP, the capital stock, and liquidity. The lower the capital stock, the higher the returns on capital. Therefore, when capital is low, savers have a high demand for investment, so for a fixed level of liquidity, the price of capital must be high to clear the market. Similarly, Figure 3 shows that the higher the productivity, the higher the q . In contrast, when liquidity is low, holding the return to capital fixed, the supply of investment projects is scarce. This is why the panels on the left-hand side of the figure show how lower liquidity increases q . This is an important feature of the model, and we explain below how it determines risk-premium dynamics.

We next describe the policy functions for the consumption-wealth ratio. As shown in (16) and (14), these policy functions are associated with each agent's marginal valuation of capital. When the EIS is greater than one, agents save more when returns are higher—that is, the intertemporal substitution effect dominates the wealth effect. This is the case in our calibration, and this force dominates the behavior of the marginal propensity to consume. Figure 4 shows that the consumption-wealth ratio is an increasing function of capital since the return to capital is decreasing in the capital stock³. The same is true when productivity is relatively low: Returns are low, so the propensity to consume is high. However, the effect of liquidity is different. When liquidity is abundant, the price of equity is low. In that case, savers and investors feature similar consumption-wealth ratios. In contrast, when liquidity is scarce, investors and savers face different effective prices of capital. For the saver, capital is expensive and he faces low returns; as a result, he consumes a higher portion of his wealth. For the investor, on the other hand, this is time to

³I.e., decreasing returns to scale in capital.

increase their investment rate and exploit their arbitrage opportunities.⁴

We now study investment dynamics. Figure 5 displays the investment rate (I/Y) throughout the state space. As we did with the previous policy functions, we first observe that the investment rate is decreasing in the capital stock dimension. This feature depends on the EIS: Under the calibration in which $EIS > 1$, agents invest at lower rates when returns are lower. (We extend this logic below, when we analyze the role of the EIS in the model.) On the other hand, both productivity and liquidity operate in the intuitive direction: The lower the productivity/liquidity the lower the investment rate. Thus, negative liquidity shocks depress investment rates. However, we can readily observe from this figure that the effect of a liquidity shock is not quantitatively important. We explore this feature further when we study the impulse-response dynamics of a liquidity shock.

Equity returns and liquidity. Next, we focus on equity return dynamics. In particular, we study the return on equity and its decomposition into a market return, a liquidity factor, and an amplification term. We start by analyzing the market return on equity—namely $E_t [(r_{t+1} + \lambda q_{t+1}) / q_t]$. By constructing this return, we can study the equity premium in the KM model and contrast it with previous asset-pricing literature.

Expected returns are the sum of expected dividend yield, $E_t [r_{t+1} / q_t]$, and expected capital gains, $\lambda E_t [q_{t+1} / q_t]$. Figure 6 displays these three objects. With respect to capital, there is a nonmonotonic relationship in total expected returns on equity. In principle, both the dividend yield and capital gains could explain this nonmonotonic relationship, because both $E_t [r_{t+1}]$ and $E_t [q_t]$ are decreasing in the capital stock—yet, the numerator term $E_t [r_{t+1}]$ dominates.⁵ Hence, the overall nonmonotonic relationship between total expected returns and capital is explained by capital gains. This is because as the economy accumulates capital, the liquidity constraint is less likely to bind, and hence q_t approaches 1. Therefore, the term $E_t [q_{t+1} / q_t]$ increases, because prices can only increase as $q_t \downarrow 1$. In other words, the direction of future change in prices is only upwards; q is below its mean and expected to increase. In contrast, when capital is low, the price of equity is high. However, the economy is expected to accumulate capital, and thus, its price is expected to fall. Figure 6 also shows that expected returns are lower when productivity is higher. This effect is unambiguous: It appears in both expected capital gains and dividend yields. But this nonmonotonic relationship of expected capital gains with respect to capital depends on the level of productivity. This is simply because different levels of productivity affect the probability that constraints bind, which ultimately dictates future changes in q . Notice that although expected returns can exhibit negative values throughout the state space, the unconditional mean is 3.7% in annual terms.⁶

Importantly, liquidity affects expected capital gains—and therefore the expected return on equity. In particular, the higher the liquidity, the higher the expected capital gains. This feature is a key counterfactual element in the model: We usually think of periods of abundant liquidity as tranquil times, during which expected returns on risky assets are low. This is not the working logic in the KM model, however, in which tranquil times are associated with lower levels of q —and thus expected changes in q are higher. This

⁴There is a point at which productivity is sufficiently low, returns on equity are sufficiently low and the economy operates under slack constraints. At such a point—and at lower levels of productivity—consumption policies are equivalent for savers and investors.

⁵Dividend yields are decreasing in capital, simply due to decreasing returns to scale in the stock of capital.

⁶We elaborate on the quantitative properties below.

is a crucial point in this model, which also dominates equity premium dynamics, as we discuss below.

We next discuss the liquidity component, $E_t [l_{t+1}/q_t^R]$, of returns and the amplification (η_t) that arise in an investor's return on wealth, as demonstrated in proposition 4 (equations 19 and 20). Figure 7 shows that the liquidity component is decreasing in both the stock of capital and in the level of productivity. As mentioned above, this last feature is in line with the logic proposed by Bigio (2015) and Kurlat (2013). We find that, conditional on a level of aggregate liquidity, the liquidity component of returns is smaller (in absolute value) when productivity is low. However, notice that this component is relatively stable across the state space, as opposed to the “market” component described above. Also, the wedge is decreasing in capital due to decreasing returns to scale: The higher the level of capital, the lower the returns on capital; this implies that endogenous responses to the price of equity are less pronounced. Thus, the endogenous wedge between q_t and q_t^R is smaller, which decreases $E_t [l_{t+1}]$.

Lastly, Figure 8 shows the amplification term $\eta_t = q_t/q_t^R$. This term is useful, because it illustrates the segments of the state space where the economy is constrained. The intuition for the behavior of η_t is very similar than to the intuition for q_t since q_t ultimately characterizes the behavior of q_t^R —and therefore the ratio q_t/q_t^R . Thus, the idea of η_t being decreasing in capital is also similar; again, it follows due to decreasing returns to scale—i.e., returns on capital are lower when the level of capital is higher. Also, by definition, the economy is unconstrained in states in which $\eta_t = 1$ —i.e., when $q_t = 1$ —, and the level of $\eta_t > 1$ gives a sense of proportion between q_t and q_t^R . In fact, this proportion depends on the value of θ , the parameter that governs the enforcement constraint. The kink observed in the right panels of Figure 8 is the precise point at which the constraint binds. Observing this feature is one of the advantages of solving the model using global solution methods and of understanding the state space of an economy with occasionally binding constraints—as in recent macro models such as Brunnermeier and Sannikov (2014) and He and Krishnamurthy (2013).

The model does not explicitly introduce a one-period risk-free bond. However, we can price such an asset by using the savers' stochastic discount factor, which would be the marginal agent pricing it⁷

$$\frac{1}{R_{f,t}} = E_t [M_{t,t+1}^s]. \quad (22)$$

Figure 9 shows that liquidity has overall negligible effects on the implied real interest. However, it does fluctuate with capital and productivity. Those fluctuations are similar to that of the neoclassical framework. That is, when the level of capital is high, the economy is expected to have a lower growth rate. Lower expected growth in consumption implies that real rates are lower in order to clear the goods market. Productivity level is simply a scaling effect for this intuition: The higher the productivity level, the higher the growth rate—and therefore also the risk-free interest rate. As shown in the figure, interest rates could be negative when productivity is low. However, the unconditional mean is 1.7% in annual terms, as we report in Table 2 and 3 (described below).

⁷Investors are not the marginal agent who prices a riskless asset, because they have the opportunity to invest in capital. Thus, $E[M^s] > E[M^i]$ always.

Lastly, we compute the (log) equity premium to evaluate whether the model can reproduce sizable premiums, in line with the empirical evidence. In particular, we compute

$$rp_t = E_t \left[\log \left(\frac{r_{t+1} + \lambda q_{t+1}}{q_t} \right) - \log (R_{f,t}) \right]. \quad (23)$$

We express the premium in logs to make it comparable with the literature (see below), and present the results in figure 10. The equity premium inherits the behavior of expected returns—which, as explained above, are dictated by expected capital gains in the price of equity. The evidence suggests that premiums are usually countercyclical: high in bad times and low in good times. In Figure 10, we observe that premiums obey this logic with respect to capital and productivity. In other words, in good time—i.e., high levels of capital and productivity—expected returns are low. This is not the case, however, for liquidity. As mentioned above, higher levels of aggregate liquidity imply lower prices and higher expected capital gains and, therefore, premiums. Interestingly, the figure shows the nonmonotonic relationship around the point at which constraints binds. This is a particular feature of this model relative to the frictionless economy. A similar body of literature that studies how balance-sheet dynamics affect risk premia also predicts a non-monotonic relationship between excess returns and the distribution of wealth.⁸ This feature allows the framework to capture the substantial volatility of excess returns—a feature well documented in the data and one that the standard frictionless RBC economy cannot account for.

At this point, it is useful to further study the role of ψ and its implications over excess returns. This is useful, because in leading asset-pricing models the signs of risk prices are quite sensible to the parametrization of ψ (all other things equal⁹). When the EIS is less than one, the consumption-wealth ratio is a decreasing function of capital. This translates into higher investment rates in periods when capital is high. Thus, on average, prices for equity are higher, and constraints are more likely to bind. For example, with $\psi=0.5$ —and keeping other parameters equal—constraints are very likely to bind throughout the state space. That is, $q > 1$ almost always. Also, as is well known, the lower the EIS ψ , the higher the mean risk-free rate.

Furthermore, when the EIS is lower than one, the equity premium is higher the greater the level of capital. This is simply because the sensitivity of the interest rate with respect to the capital stock is higher, and therefore an increase in the capital stock reduces the risk-free interest rate relatively more than the expected return on equity. Thus, expected excess returns are higher. It is worth emphasizing that the relationship with respect to liquidity and productivity is the same as in the case of ψ greater than one.

In conclusion, the model offers a counter-factual result in terms of the impact of liquidity on expected returns: Liquidity carries the wrong sign in the price of liquidity risks. This result is independent of the EIS, and it fundamentally depends on expected fluctuations in the price of equity. The model can reproduce a higher mean excess return and a considerably higher volatility relative to the frictionless benchmark. We next assess the quantitative properties of business cycles.

⁸Brunnermeier and Sannikov (2016) provide a comprehensive review.

⁹For example, Bansal and Yaron (2004) show that when ψ is less than one, the price of volatility shocks changes the sign.

Business-cycle properties. We now study the quantitative properties of the business cycle predicted by the model. To do so, we follow both the RBC literature and the production-based asset pricing literature—Jermann (1998), Campanale et al. (2010), Croce (2014), among others—and compute key asset pricing and business cycle moments.

[Table 2 about here: Asset prices and business cycles]

Table 2 displays the results and the comparison to two other well-known models in the production-based asset pricing literature. The model predicts fluctuations of investment almost 2.5 times more volatile than output, which is much more than the 1.46 predicted by the frictionless RBC. This is not that far from what we observe in the data, especially taking into account that our model does not exhibit adjustment costs to investment, as in Jermann (1998) or Croce (2014). It is well known that this statistic could be easily matched by adding such costs to investment, but we purposefully abstract from those to focus strictly on the q -theory mechanism embedded in the KM framework.

We also see that the unconditional mean for the risk-free interest rate is above the one observed in the data, but definitely smaller than the one predicted by the frictionless benchmark.¹⁰ In the other models we compare with, the low interest rates is driven by the introduction of a small persistent component into the expected growth rates (Croce (2014)) or the calibrated process for the habits in preferences (Jermann (1998)).

The unconditional expected excess return on equity is 1.05% per annum. This magnitude is considerably smaller than the one observed in the data. However, the model predicts a sizable amount of variability in equity returns—and thus in expected excess returns. Although the standard deviation we obtain is not as high as in the data, it is a considerable improvement relative to the frictionless case in which variability of both returns and expected returns is very small. Although the model predicts substantial variation, it is important to bear in mind that the directions of such fluctuations are counterfactual to the evidence, as explained in our description of expected excess returns over the state space.

Lastly, in Figure 2, we show the empirical distribution of excess returns, both in the data and in the model. This figure complements Table 2. It is evident that excess returns in the data display larger fluctuations than in the model; that is, excess returns are more volatile than in the model. One potential avenue to address this is to introduce exogenous second-order shocks to productivity (see Schneider (2017)). In this way, fluctuations in premiums would be attributed in part to the endogenous component driven by occasionally binding constraints, and also to exogenous second order shocks. It is useful to stress that the case of the frictionless case RBC predicts a degenerate distribution very close to zero.

Decomposition. We next study the decomposition of returns. As established by (19) and (20), returns on wealth can be decomposed into three terms: an intercept (or liquidity component), a slope (or amplification component) and the market return on equity. In Table 3, we report the moments for each of these variables. We use the expression $E[E_t(x_{t+1})]$ for a given variable x because we first compute the con-

¹⁰Although we could reduce the mean risk free rate by increasing β , we keep it at 0.99 to improve the speed of convergence of the algorithm.

ditional expectation across the state space, using the Markov transition probabilities, and then compute the unconditional expectation using the corresponding distribution—i.e. unconditional or conditional distribution.¹¹ The appendix contains a detailed description of how we compute the unconditional and conditional distributions.

[Table 3 about here: Decomposition]

As shown in Table 3, the compensation for liquidity is negative, and relatively smaller than the pure market return on equity (in absolute terms). Intuitively, this is because there is a strictly positive probability (π) that a fraction $(1 - \phi)$ of the equity will be valued at a price (q^R) that is inferior to that of the spot price (q). This is always the case, since there is always a probability of switching between constrained and unconstrained parts of the state space. In fact, the system spends 85% of the time in the constrained part of the state space.

Table 3 illustrates the key counter-factual element of the model: The conditionally constrained expected return on equity is considerably lower than the conditionally unconstrained one. This is counterfactual, because we usually think that constraints are slack in tranquil times—i.e., when premiums are low. However, the volatility of equity returns is higher in the constrained part of the state space. Thus, when financial constraints bind, returns on equity are on average more volatile. This last aspect of the model is promising, since it can potentially contribute to explaining the observed large fluctuations in excess returns.

The average effect of liquidity is notably similar across the state space—i.e., it does not change significantly between constrained and unconstrained state space. Qualitatively, both $E[E_t(l_{t+1})]/q_t^R$ and $E[E_t(l_{t+1})]/q_t$ are on average higher, in absolute terms, when the economy is constrained. However, this difference is quantitatively mild. In other words, the contribution of liquidity to total returns on wealth is relatively small and similar across the state space.

The last element to analyze is the amplification factor η . From Table 3, we see that the unconditional η is 1.94, while the conditionally constrained mean is somewhat higher at 2.07. These values for η imply a somewhat unrealistic expected return on wealth for investors, since η multiplies the gross market return and does not affect the intercept—i.e., the liquidity component. Thus, we see that the expected return on investor's wealth is

$$E \left[\frac{E_t(l_{t+1})}{q_t^R} + \eta E_t \left(\frac{r_{t+1} + q_{t+1}}{q_t} \right) \right] = 1.95$$

Although we don't have benchmark data to compare this number with, it seems to be relatively large. This is because η_t works as an amplification of gross returns.

Impulse response functions. We next examine the economy's response to a liquidity shock. We compute impulse-response functions (IRFs) using a simulation algorithm described in the appendix. In Figure 12, we show the evolution of the economy after a relatively big liquidity shock. The shock reduces liquidity from $\phi_t = 0.275$ —the steady-state level—, to $\phi_t = 0.05$; these results are in line with the intuition we developed above.

¹¹We would have used the law of iterated expectations if we had only included the unconditional column.

We find that the response in output is a drop of 0.12% (approximately 0.5% annualized). This is mainly explained by a drop of 9.5% in the level of investment or, equivalently a 9.8% drop in the *investment rate*: After the shock, the economy moves from a 23% investment rate to a 20.5% rate.

Both hours and wages drop, although consumption increases after the shock. The logic is that the shock affects the supply of capital, which reduces investment and, therefore, capital accumulation. Output also drops, but since we have a closed economy with flexible prices, consumption must increase to clear the goods market.

After the liquidity shock, we observe that the probability that constraints will bind increases up to one. The price of equity, q_t , increases about 15% above its mean. The expected excess return, however, drops around 10% from its mean. This is the main counterfactual element we highlighted above: Periods of scarce liquidity are associated with lower expected excess returns. As noted above, this incorrect sign in the price for liquidity risk is not associated with the entrepreneur's EIS. Instead, it is embedded in the logic that liquidity shocks affect the supply of equity without affecting its demand. As a result, the price for risky equity increases, and investors expect that it will decrease in the near future.

Theoretical moments and invariant distribution. We conclude our quantitative analysis by analyzing the theoretical moments of endogenous variables. Table 4 shows the theoretical moments. We complement this table with Figure 11. We provide a detailed discussion in the appendix of how we compute the invariant distribution, but the idea is to solve an eigenvalue problem associated with the Markov chain. In the table, we report moments computed with three measures: unconditional, conditionally constrained, and conditionally unconstrained. We observe that capital, investment and output have lower means under the conditionally constrained distribution. Also, the conditionally constrained second moments of capital, output, and investment are relatively higher than the unconditionally constrained moments. This is consistent with the intuition we elaborate on throughout the text.

[Table 4 about here: Theoretical moments]

Interestingly, the skewness and kurtosis of the endogenous variables inform us that the conditionally unconstrained distribution is negatively skewed and displays higher levels of kurtosis. The intuition is that the unconstrained state is transitory; indeed, in our calibration the economy spends 87% of the time in the constrained state. Figure 11 illustrates a similar intuition and is complementary to Table 4. At first blush, we can see that the conditionally constrained distribution assigns smaller measures to relatively higher levels of capital.

4 Conclusion

In this paper, we propose a stripped-down version of KM. Our objective is to evaluate the quantitative predictions of liquidity shocks through the lens of asset prices and business cycles. Our main conclusion is that liquidity shocks, coupled with financing constraints, are a promising avenue of research that can potentially capture highly nonlinear risk premium dynamics.

However, at the core of the KM mechanism, we find that negative liquidity shocks increase the price of equity, and this translates into risk premium dynamics. Our results complement and reinforce other findings in the literature (e.g., [Shi \(2015\)](#)). We extend our analysis to study risk premium dynamics and highlight that although fluctuations in this variable are considerably more realistic than in the frictionless benchmark, the directions of those fluctuations are at odds with the evidence.

Finally, we observe that the fact that liquidity shocks operate solely on the “supply side” is a major restriction of this setup. A potential avenue of future research could shed light on how liquidity affects the demand side of the credit market—that is, contractions in the demand for credit would be associated with higher expected excess returns and lower asset prices.

TABLE 1. Calibration

PARAMETERS (QUARTERLY)			
1. Technology	Value	Description	
α	0.36	capital share	
λ	0.975	1-depreciation	
π	0.012	investment opportunities	
θ	0.77	borrowing constraint	
2. Preferences			
β	0.99	time preference	
γ	7.5	risk aversion	
ψ	1.5	EIS	
ν	1	Frisch elasticity	
κ	8.5	scaling hours	
3. Aggregate shocks			
ρ_A	0.95	AR(1) productivity	
σ_A	0.01	standard deviation $\log(A_t)$	
φ	0.75	$\text{corr}(A, \phi)$	
ϕ_L	0.05	minimum level of liquidity	
ϕ_H	0.5	maximum level of liquidity	

NOTES: The table shows the model calibration at a quarterly frequency. We describe the procedure in the main text.

TABLE 2. Asset prices and business cycles

Model/moments	$\sigma_{\Delta c}/\sigma_{\Delta y}$	$\sigma_{\Delta i}/\sigma_{\Delta y}$	$E[\log R_{f,t}]$	$E[\log R^e - \log R_{f,t}]$	$\text{std}(\log R_{f,t})$	$\text{std}(\log R^e)$
KM	0.91	2.40	1.72	1.05	0.89	9.32
Frictionless	0.84	1.46	4.01	0.01	0.26	0.29
Croce2014	0.81	3.61	0.94	5.25	0.94	12.47
Jerman1998	0.49	2.64	0.82	6.18	11.46	19.86
Data	0.71	4.49	0.65	4.71	1.86	20.89

NOTES: This table shows the main business cycle statistics. We compute the moments by simulating the model for 500,000 periods (we burn 10,000 periods). $\sigma_{\Delta j}/\sigma_{\Delta y}$ is the standard deviation of j =consumption, investment growth over output growth. KM is the model presented in the main text. We also report theoretical moments for $E[R_f]$ and $E[R^e]$ in table 3. Frictionless is the solution when there is neither incentive compatibility nor resellability constraints. Jerman1998 is (Jermann, 1998) Table 1, Benchmark Model (habits+adjustment costs to investment). Croce14 is Croce (2014) Table 3, column 3 (EIS>1). Data is from Croce (2014), which is annual data from 1929-2008.

TABLE 3. Decomposition

<i>Variable</i>	Unconditional		Constrained		Unconstrained	
	Mean	St.dev.	Mean	St. Dev	Mean	St. dev.
$E [E_t (l_{t+1})] / q_t^R$	-0.0079	0.0084	-0.0088	0.0086	-0.0021	0.0020
$E [E_t (l_{t+1})] / q_t$	-0.0035	0.0020	-0.0037	0.0019	-0.0021	0.0019
$E \left[E_t \left(\left(\frac{r_{t+1} + q_{t+1}}{q_t} \right) \right) \right]$	1.0093	0.0507	1.0026	0.0483	1.0561	0.0416
$E [R_{f,t}]$	1.0044	0.0062	1.0040	0.0062	1.0073	0.0054
η_t	1.9406	1.1414	2.0767	1.1596	1	0

NOTES: This table shows the moments under unconditional, conditionally constrained, and conditionally unconstrained distributions of the state variables. Numbers are expressed in decimal units and in quarterly terms. We discuss in the appendix how we compute unconditional, conditionally constrained, and unconditionally constrained distributions.

TABLE 4. Theoretical moments

Variable	Moments			
	Mean	St.dev.	Skewness	Kurtosis
Panel A. Unconditional Distribution				
K_{t+1}	10.941	1.606	-0.154	1.404
A_t	1.000	0.030	0.071	2.524
ϕ_t	0.275	0.139	0	1.788
q_t	1.091	0.071	0.358	1.944
Y_t	1.051	0.098	-0.009	2.071
I_t	0.243	0.027	-0.151	2.512
Panel B. Conditionally constrained distribution				
K_{t+1}	10.735	1.591	0.053	1.405
A_t	0.997	0.031	0.131	2.658
ϕ_t	0.248	0.128	0.157	1.969
q_t	1.110	0.065	0.289	1.943
Y_t	1.035	0.090	0.035	2.100
I_t	0.239	0.025	-0.177	2.557
Panel C. Conditionally unconstrained distribution				
K_{t+1}	12.391	0.714	-2.132	7.645
A_t	1.021	0.0315	-0.641	2.997
ϕ_t	0.457	0.038	-0.547	2.423
q_t	1	0	0	0
Y_t	1.159	0.069	-1.146	4.822
I_t	0.272	0.019	-0.595	2.9157

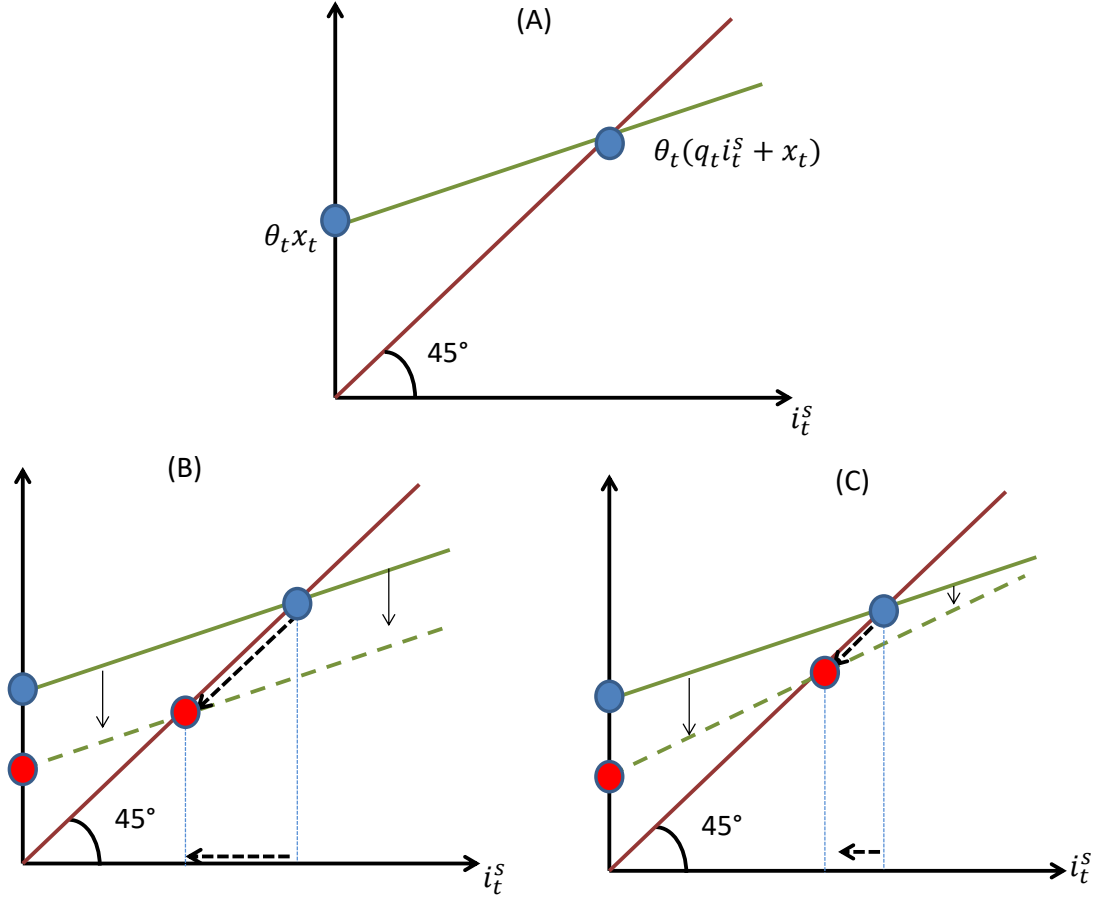
NOTES: This table reports the theoretical moments of relevant endogenous variables in the model. Panel A reports the moments computed with the unconditional distribution of the state variables. Panel B reports the moments computed with the conditionally constrained distribution (i.e., conditional on $q > 1$). Panel C Reports the moments under the conditionally unconstrained distribution (i.e., conditional on $q = 1$). We report in the appendix how we compute these distributions. Kurtosis and skewness are

$$kurt_{\pi}[z(s)] = \frac{\mathbb{E}_{\pi}[(z(s) - \mathbb{E}_{\pi}(z(s)))^4]}{(\mathbb{E}_{\pi}[(z(s) - \mathbb{E}_{\pi}(z(s)))^2])^2}$$

$$skew_{\pi}[z(s)] = \frac{\mathbb{E}_{\pi}[(z(s) - \mathbb{E}_{\pi}(z(s)))^3]}{(\mathbb{E}_{\pi}[(z(s) - \mathbb{E}_{\pi}(z(s)))^2])^{3/2}}$$

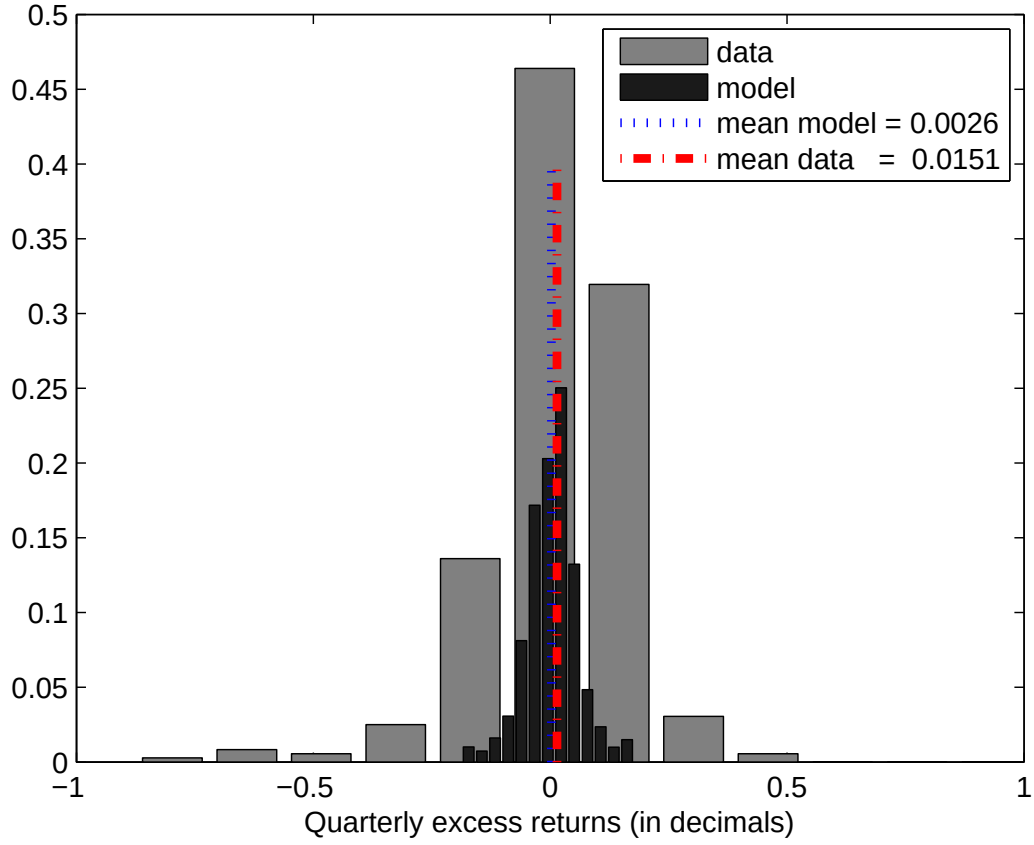
where π is the corresponding measure (unconditional, conditionally constrained, or conditionally unconstrained).

FIGURE 1. Intuition behind the effects



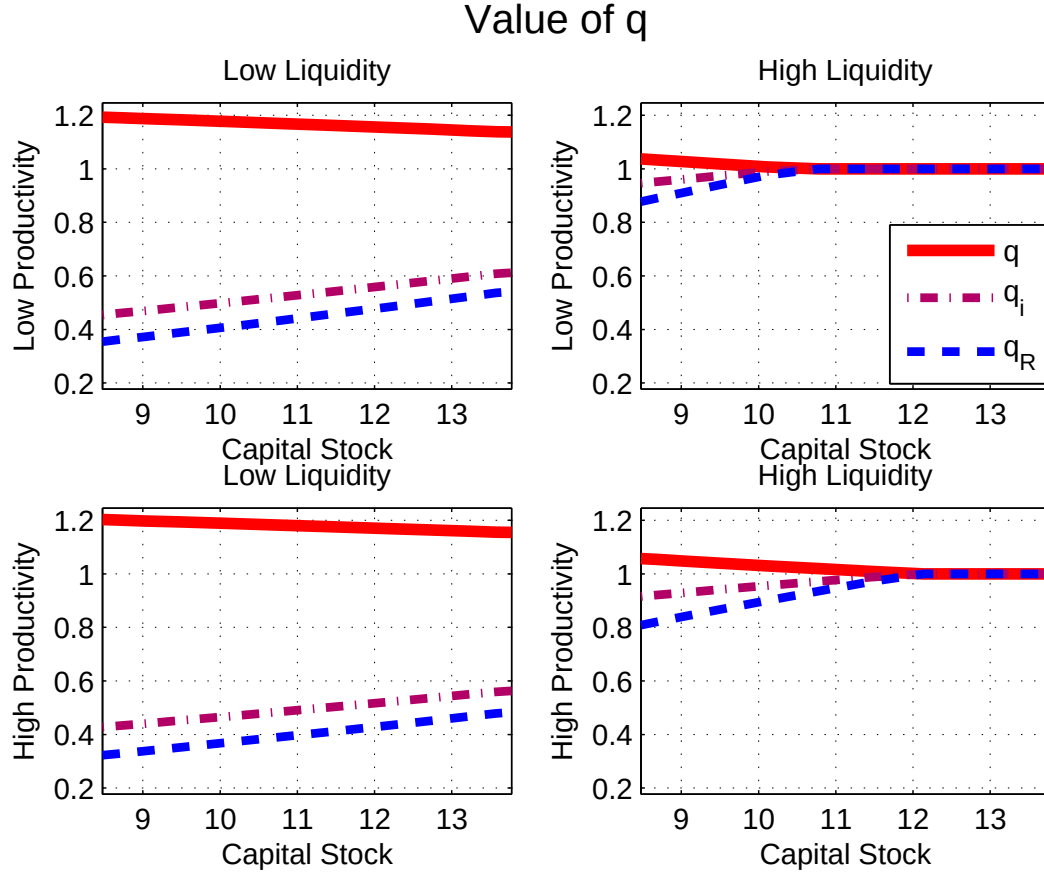
NOTES: Panel (A) shows how borrowing constraints impose a cap on the amount of equity that can be sold to finance a downpayment. Panel (B) shows how liquidity shocks affect the amount of resources available as a down payment. Panel (C) shows the price effect of the shock, which may or may not reinforce the liquidity shock.

FIGURE 2. Empirical distribution of excess returns



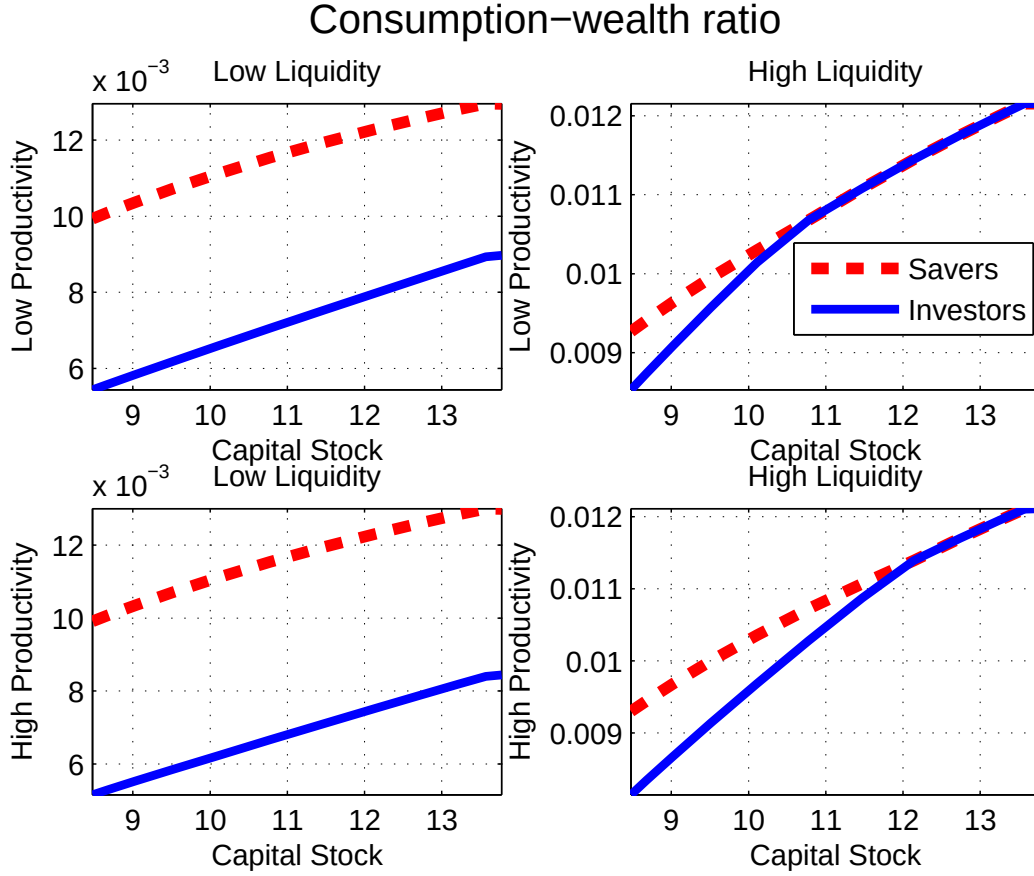
NOTES: This figure shows the empirical distribution for the risk premium(i.e., equation 10) in the data and in the model. The model distribution is obtained by simulating the model 500,000 periods (we burn the first 10,000 periods). Data are constructed exactly as in Croce (2014), but at a quarterly frequency and with an extended time window (1927q1-2016q4 instead of 1929-2004). For the time interval considered in Croce (2014) and reported in table 2, the data used in this figure yield very similar numbers: $E[\log R_{f,t}] = 0.62\%$, $E[\log R^e - \log R_{f,t}] = 5.5\%$, $std(\log R_{f,t}) = 1.93\%$, $std(\log R^e) = 19.55\%$, all annualized.

FIGURE 3. Price of equity, q_t



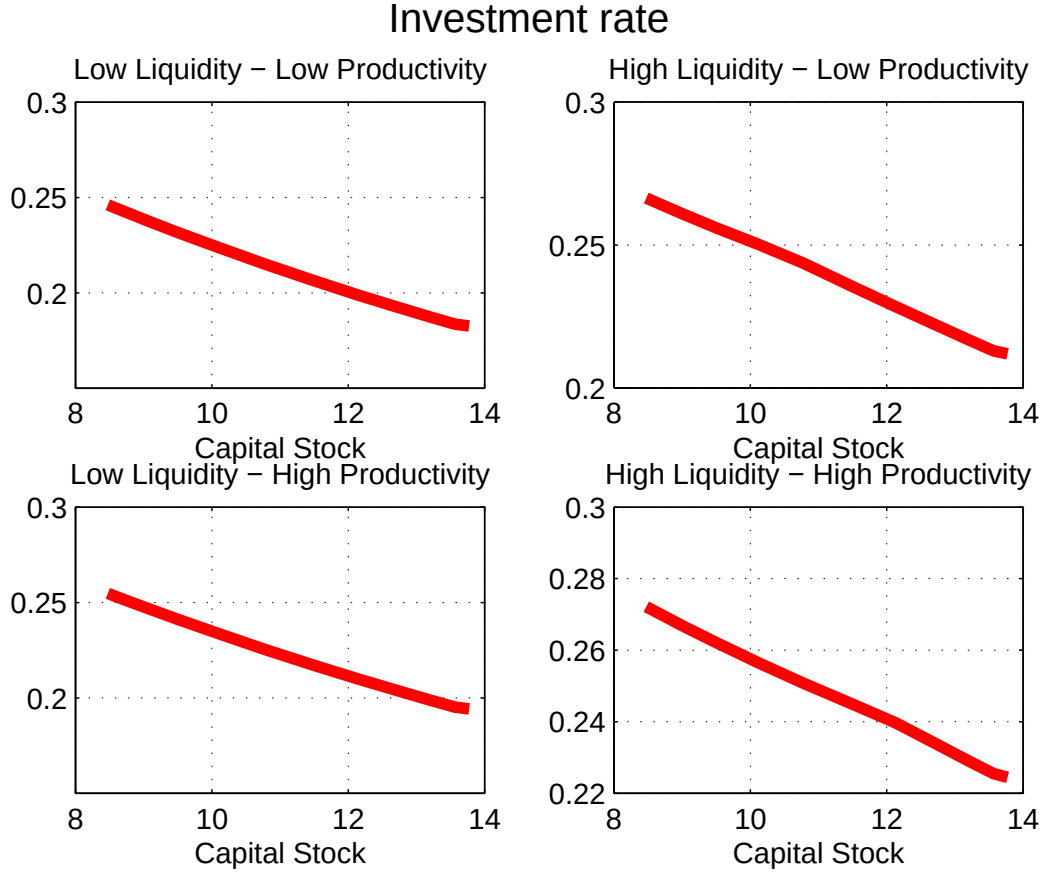
NOTES: This figure displays q across the state space. The upper-left panel represents the variable when liquidity and productivity are at their 20% percentiles. The upper-right panel represents the variable when liquidity is at its 80% percentile and productivity is at its 20% percentile. The lower-right panel represents the variable when liquidity is at its 20% percentile and productivity is at its 80% percentile. The upper-left panel represents the variable when liquidity and productivity are both at their 80% percentiles. Percentile values are $A^{(20)} = 0.97$, $A^{(80)} = 1.03$, $\phi^{(20)} = 0.1192$, $\phi^{(80)} = 0.4308$. In all subplots, the x-axis represents the level of the endogenous state variable, K . We provide details about the statistical properties of K in Table 3.

FIGURE 4. Consumption-wealth ratio

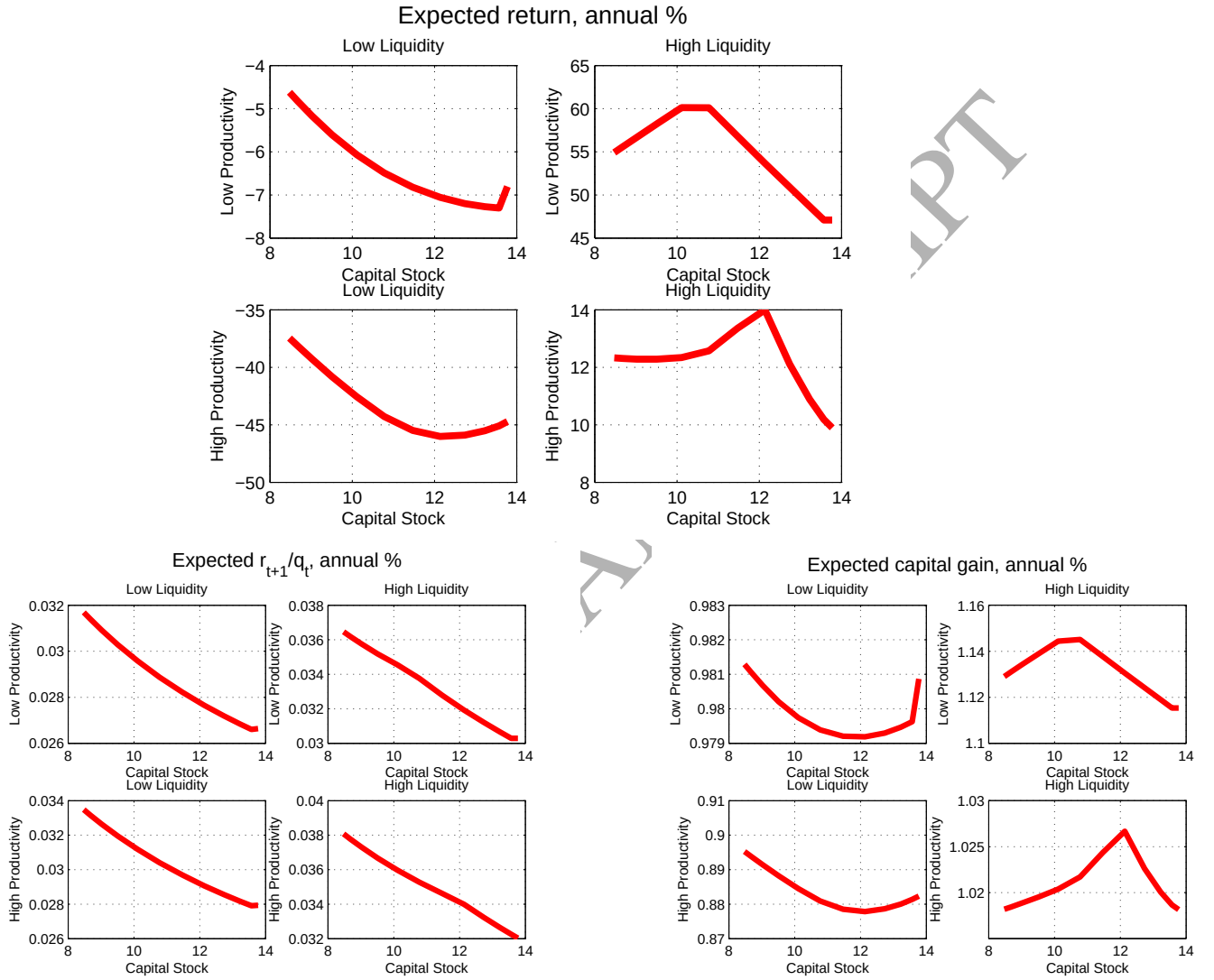


NOTES: This figure displays the consumption-wealth ratio across the state space. The upper-left panel represents the variable when liquidity and productivity are at their 20% percentiles. The upper-right panel represents the variable when liquidity is at its 80% percentile and productivity is at its 20% percentile. The lower-right panel represents the variable when liquidity is at its 20% percentile and productivity is at its 80% percentile. The upper-left panel represents the variable when liquidity and productivity are both at their 80% percentiles. Percentile values are $A^{(20)} = 0.97$, $A^{(80)} = 1.03$, $\phi^{(20)} = 0.1192$, $\phi^{(80)} = 0.4308$. In all subplots, the x-axis represents the level of the endogenous state variable, K . We provide details about the statistical properties of K in Table 3.

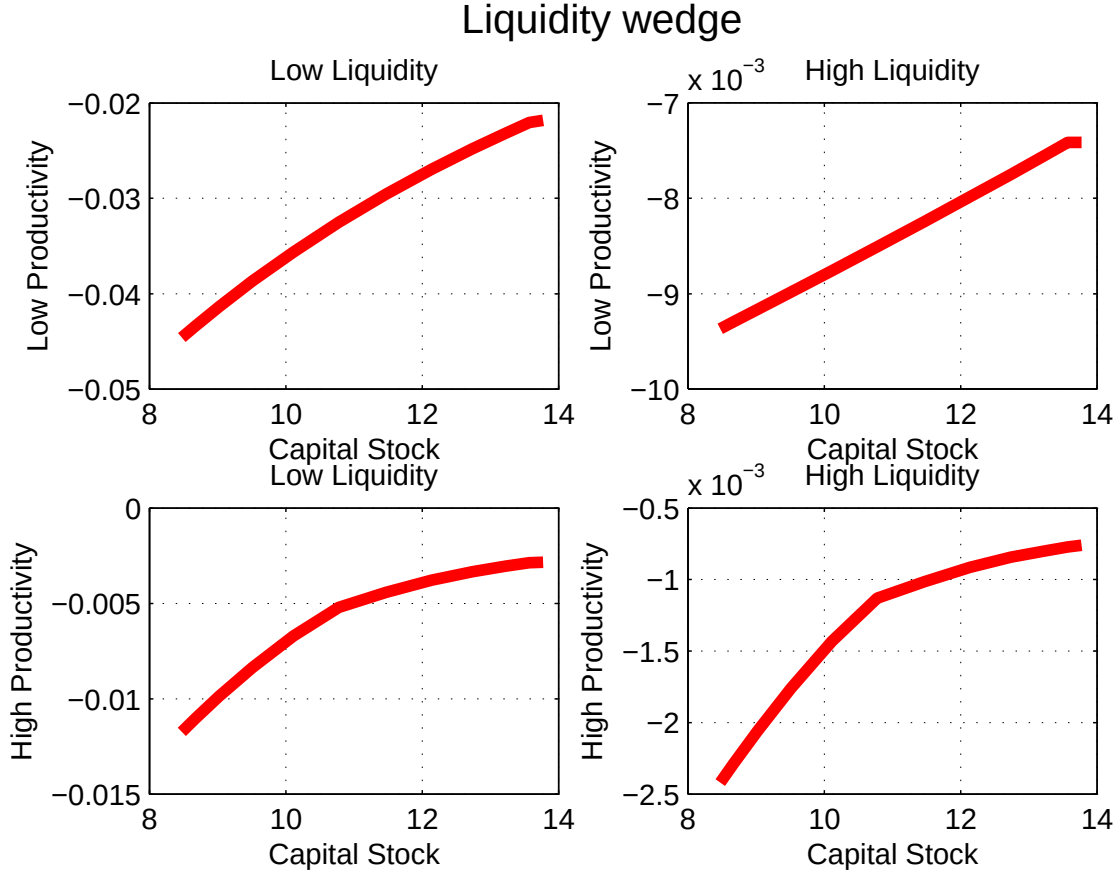
FIGURE 5. Investment rate I/Y



NOTES: This figure displays the investment rate I/Y across the state space. The upper-left panel represents the variable when liquidity and productivity are at their 20% percentiles. The upper-right panel represents the variable when liquidity is at its 80% percentile and productivity is at its 20% percentile. The lower-right panel represents the variable when liquidity is at its 20% percentile and productivity is at its 80% percentile. The upper-left panel represents the variable when liquidity and productivity are both at their 80% percentiles. Percentile values are $A^{(20)} = 0.97$, $A^{(80)} = 1.03$, $\phi^{(20)} = 0.1192$, $\phi^{(80)} = 0.4308$. In all subplots, the x-axis represents the level of the endogenous state variable, K . We provide details about the statistical properties of K in Table 3.

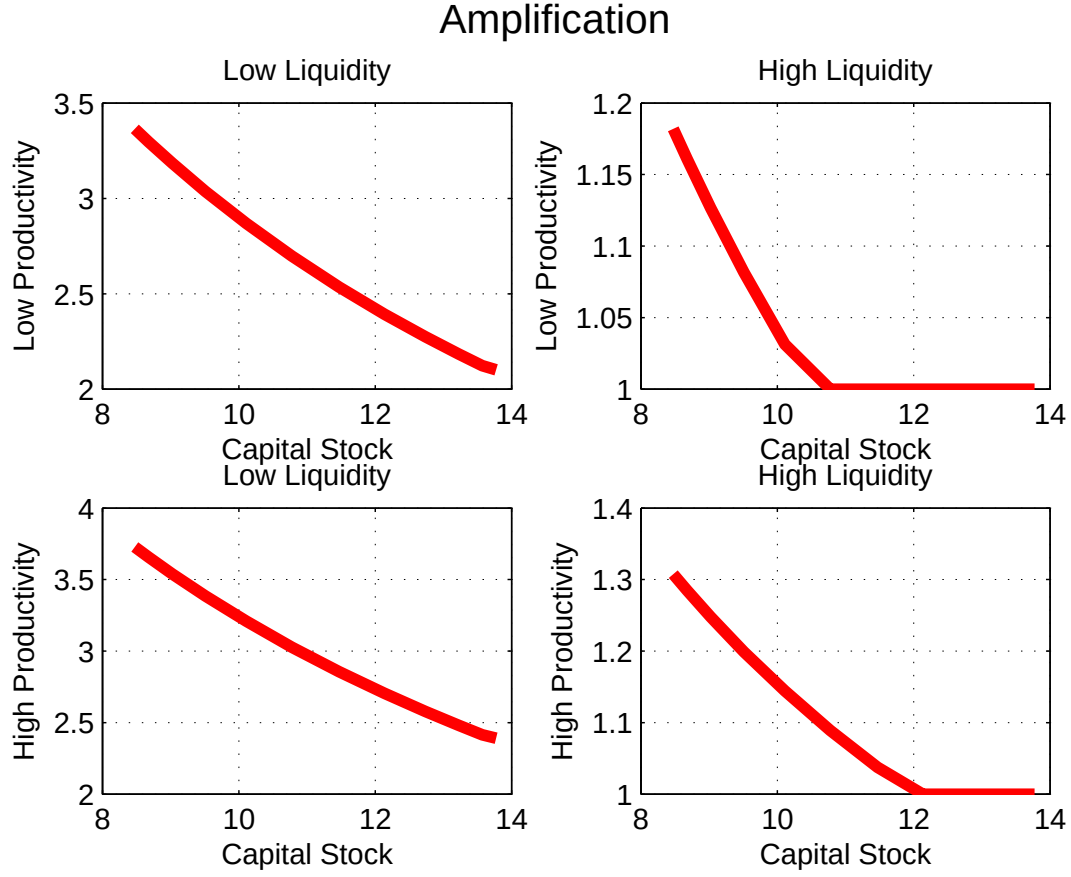
FIGURE 6. Expected return $E[(r_{t+1} + \lambda q_{t+1}) / q_t]$, capital gains and dividend yield

NOTES: This figure displays the market return on equity, the dividend yield and the capital gains across the state space. The upper-left panel represents the variable when liquidity and productivity are at their 20% percentiles. The upper-right panel represents the variable when liquidity is at its 80% percentile and productivity is at its 20% percentile. The lower-right panel represents the variable when liquidity is at its 20% percentile and productivity is at its 80% percentile. The upper-left panel represents the variable when liquidity and productivity are both at their 80% percentiles. Percentile values are $A^{(20)} = 0.97$, $A^{(80)} = 1.03$, $\phi^{(20)} = 0.1192$, $\phi^{(80)} = 0.4308$. In all subplots, the x-axis represents the level of the endogenous state variable, K . We provide details about the statistical properties of K in Table 3.

FIGURE 7. Liquidity component $E_t [l_{t+1}] / q_t^R$ 

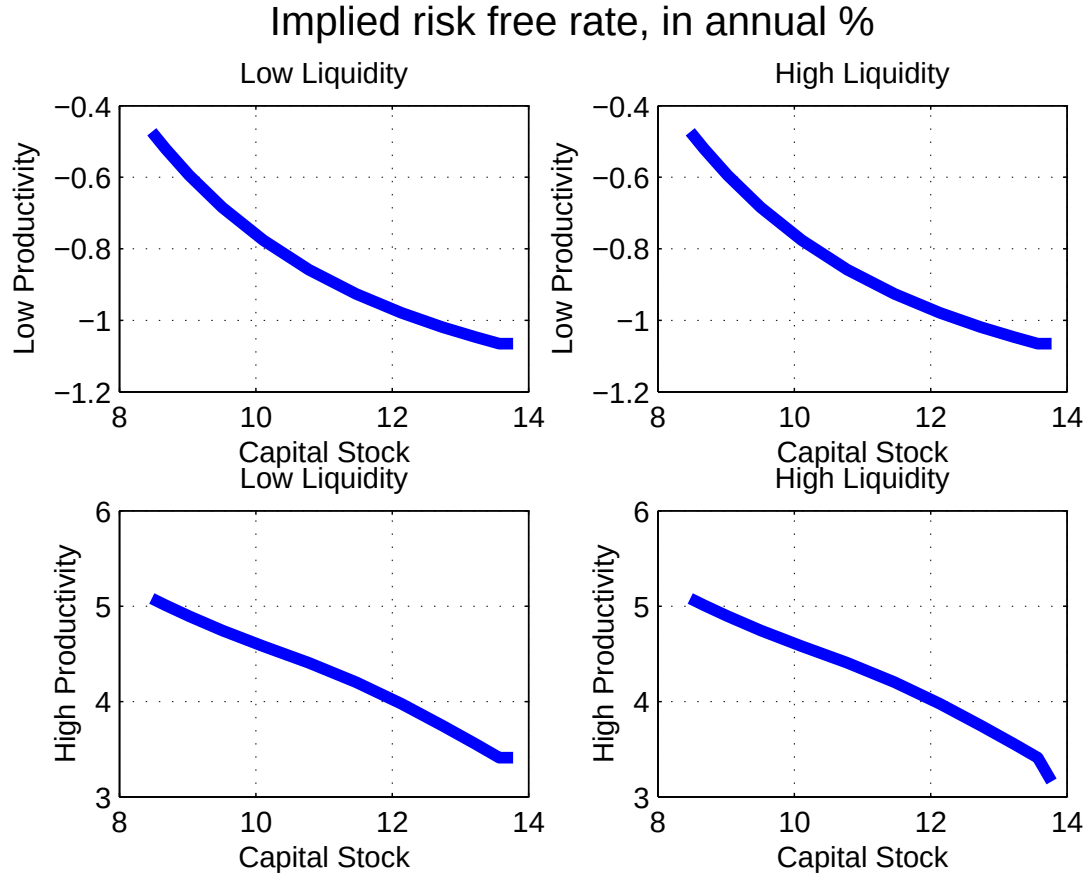
NOTES: This figure displays the liquidity component across the state space. The upper-left panel represents the variable when liquidity and productivity are at their 20% percentiles. The upper-right panel represents the variable when liquidity is at its 80% percentile and productivity is at its 20% percentile. The lower-left panel represents the variable when liquidity is at its 20% percentile and productivity is at its 80% percentile. The upper-right panel represents the variable when liquidity and productivity are both at their 80% percentiles. Percentile values are $A^{(20)} = 0.97$, $A^{(80)} = 1.03$, $\phi^{(20)} = 0.1192$, $\phi^{(80)} = 0.4308$. In all subplots, the x-axis represents the level of the endogenous state variable, K . We provide details about the statistical properties of K in Table 3.

FIGURE 8. Amplification, η_t



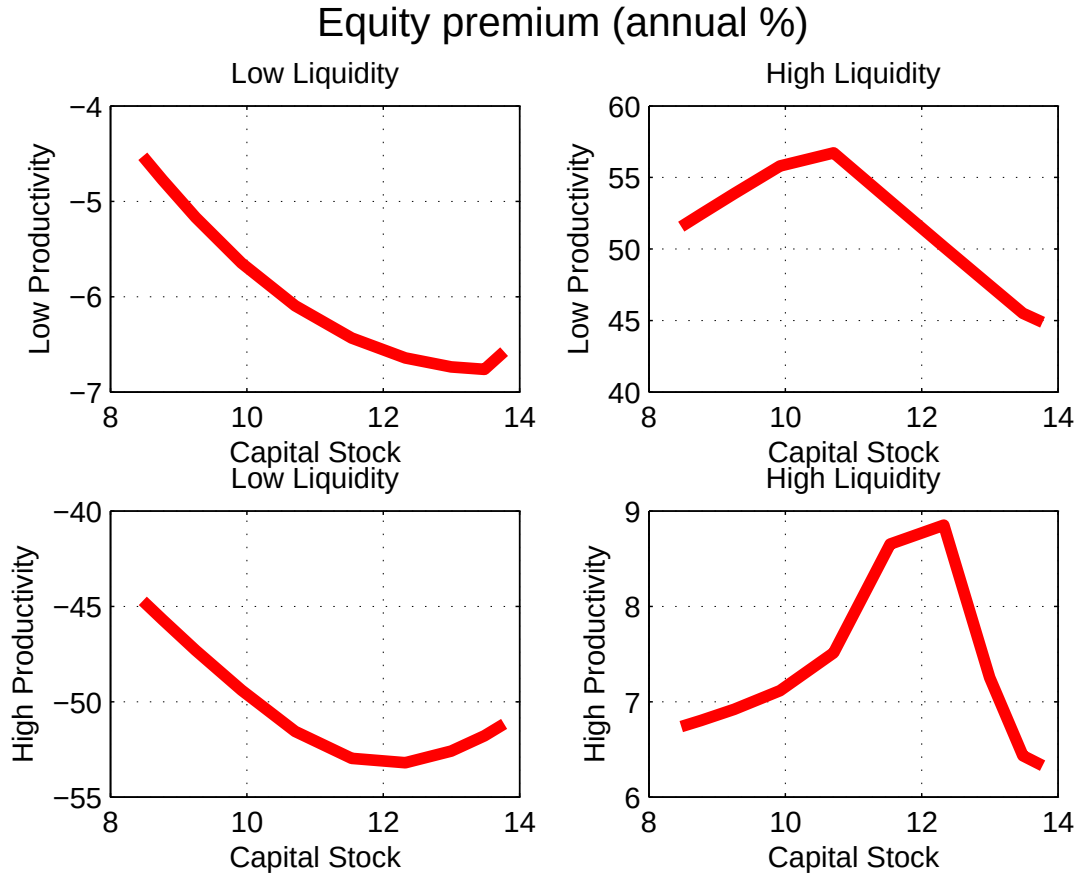
NOTES: This figure displays amplification term across the state space. The upper-left panel represents the variable when liquidity and productivity are at their 20% percentiles. The upper-right panel represents the variable when liquidity is at its 80% percentile and productivity is at its 20% percentile. The lower-right panel represents the variable when liquidity is at its 20% percentile and productivity is at its 80% percentile. The upper-left panel represents the variable when liquidity and productivity are both at their 80% percentiles. Percentile values are $A^{(20)} = 0.97$, $A^{(80)} = 1.03$, $\phi^{(20)} = 0.1192$, $\phi^{(80)} = 0.4308$. In all subplots, the x-axis represents the level of the endogenous state variable, K . We provide details about the statistical properties of K in Table 3.

FIGURE 9. Real interest rate



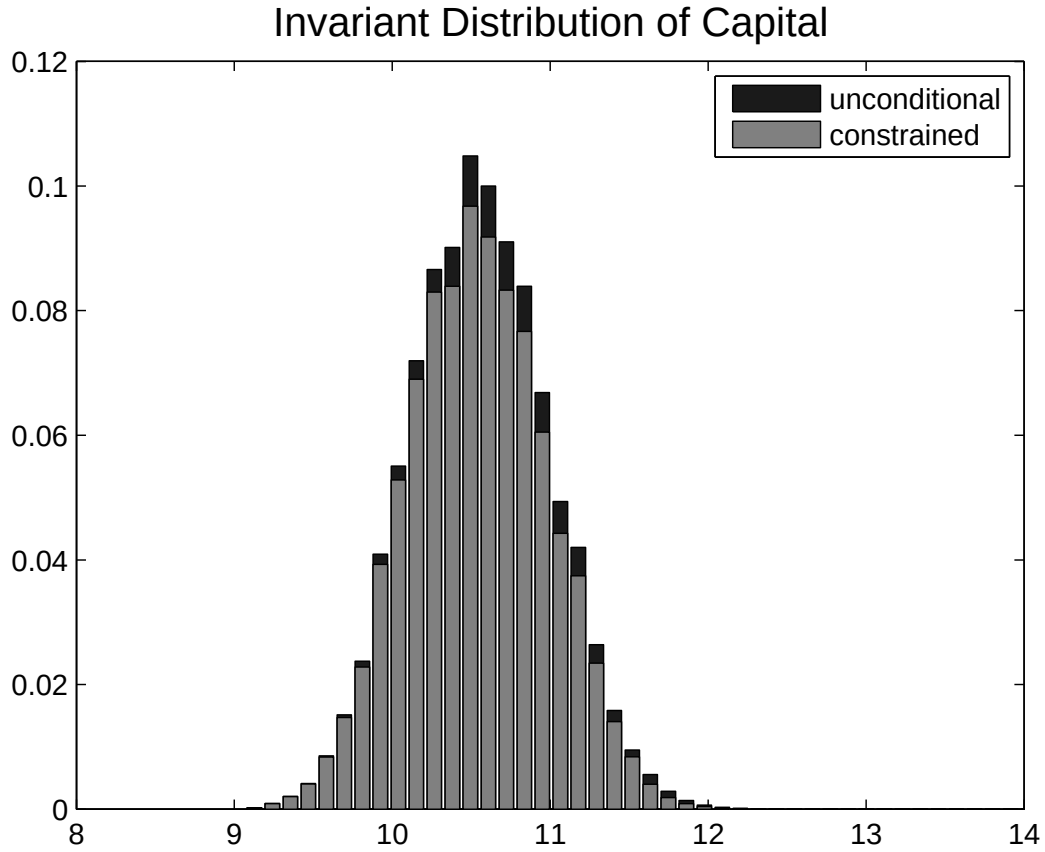
NOTES: This figure displays the real risk-free interest rate across the state space. The upper-left panel represents the variable when liquidity and productivity are at their 20% percentiles. The upper-right panel represents the variable when liquidity is at its 80% percentile and productivity is at its 20% percentile. The lower-right panel represents the variable when liquidity is at its 20% percentile and productivity is at its 80% percentile. The upper-left panel represents the variable when liquidity and productivity are both at their 80% percentiles. Percentile values are $A^{(20)} = 0.97$, $A^{(80)} = 1.03$, $\phi^{(20)} = 0.1192$, $\phi^{(80)} = 0.4308$. In all subplots, the x-axis represents the level of the endogenous state variable, K . We provide details about the statistical properties of K in Table 3.

FIGURE 10. Equity premium



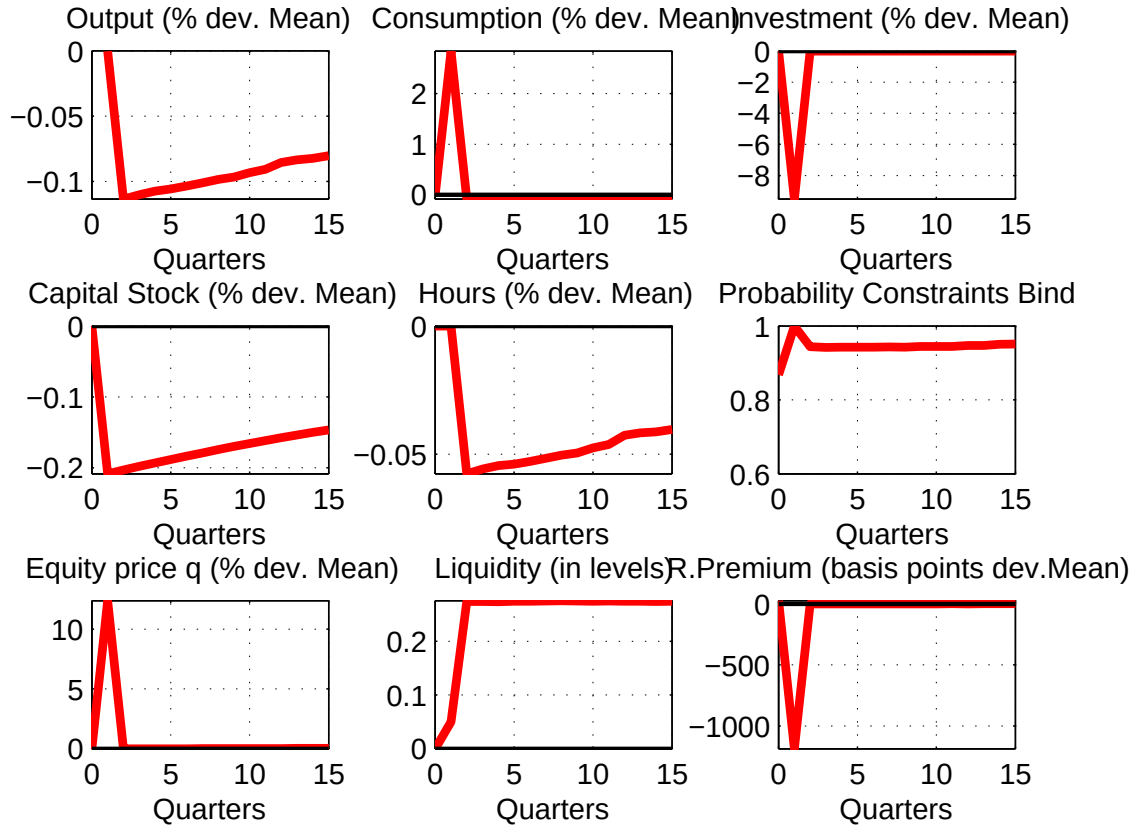
NOTES: This figure displays the equity premium across the state space. The upper-left panel represents the variable when liquidity and productivity are at their 20% percentiles. The upper-right panel represents the variable when liquidity is at its 80% percentile and productivity is at its 20% percentile. The lower-left panel represents the variable when liquidity is at its 20% percentile and productivity is at its 80% percentile. The upper-right panel represents the variable when liquidity and productivity are both at their 80% percentiles. Percentile values are $A^{(20)} = 0.97$, $A^{(80)} = 1.03$, $\phi^{(20)} = 0.1192$, $\phi^{(80)} = 0.4308$. In all subplots, the x-axis represents the level of the endogenous state variable, K . We provide details about the statistical properties of K in Table 3.

FIGURE 11. Invariant distribution of capital



NOTES: This figure shows the distribution of the endogenous state variable. The x-axis represents the level of capital and the y-axis the measure. The gray area is the conditionally constrained distribution (i.e., when constraints bind), while the black area represents the unconditional distribution. We provide the details on the construction of these distributions in the appendix.

FIGURE 12. Impulse-response functions



NOTES: This figure shows the impulse response functions after a liquidity shock. As shown in the figure, liquidity moves from its mean value of 0.275 to 0.05. We compute the response of the economy with 60,000 Monte carlo simulations, starting from the unconditional expectation of the state vector.

5 Bibliography

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6 Appendix

Proof of proposition 1 (Investor's constraint)

We need to show that the budget constraint for investors, when constraints binds, is given by

$$c_{i,t} + q_t^R n_{t+1} = (r_t + \lambda q_t^i) n_t, \quad (24)$$

where $q_t^i = \phi_t q_t + (1 - \phi_t) q_t^R$. As shown in 3, the entrepreneur's budget constraint is

$$c_t + i_t^d = r_t n_t + q_t (\Delta e_{t+1}^- - \Delta e_{t+1}^+).$$

Therefore, when constraints bind, we use $(\Delta e_{t+1}^- - \Delta e_{t+1}^+) = \lambda \phi_t n_t$

$$c_t + i_t^d = (r_t + q_t \lambda \phi_t) n_t.$$

Also

$$\begin{aligned} i_t^d &= (1 - q_t \theta) i_t, \\ &= (i_t - i_t^s) q_t^R, \end{aligned}$$

So, using the law of motion for capital

$$\begin{aligned} c_t + q_t^R i_t - q_t^R i_t^s &= (r_t + q_t \lambda \phi_t) n_t, \\ c_t + (k_{t+1} - \lambda k_t) q_t^R - q_t^R i_t^s &= (r_t + q_t \lambda \phi_t) n_t. \end{aligned}$$

Now, using the identity, $k_{t+1} = n_{t+1} - e_{t+1}^+ + e_{t+1}^-$, we have

$$c_t + n_{t+1} q_t^R + (-e_{t+1}^+ + e_{t+1}^- - i_t^s) q_t^R - q_t^R \lambda k_t = (r_t + q_t \lambda \phi_t) n_t.$$

Lastly, we use $e_{t+1}^- = \lambda e_t^- + \Delta e_{t+1}^- + i_t^s$ and obtain the desired result (24). The computation for saver follows the exact same steps. $\square$

Proof of proposition 3 (Policy functions). We first establish the solution for the value functions. Recall that the value function for agent j has the form

$$V_{j,t} = \left[c_{j,t}^\rho + \beta E \left[V_{j,t+1}^{1-\gamma} \right]^{\frac{\rho}{1-\gamma}} \right]^{\frac{1}{\rho}}.$$

We do a monotone transformation to ease the algebra

$$V_{j,t} = \left(\rho v_{j,t} \right)^{\frac{1}{\rho}},$$

so

$$v_{j,t} = \frac{c_{j,t}^\rho}{\rho} + \beta E \left[v_{j,t+1}^{\frac{(1-\gamma)}{\rho}} \right].$$

The intertemporal marginal rate of substitution is

$$\begin{aligned} M_{j,t+1} &= \frac{\frac{\partial v_{j,t}}{\partial c_{j,t+1}}}{\frac{\partial v_{j,t}}{\partial c_{j,t}}}, \\ &= \beta \left(\frac{c_{j,t+1}}{c_{j,t}} \right)^{\rho-1} \left(\frac{v_{j,t}^{\frac{1}{\rho}}}{E_t \left[v_{j,t+1}^{\frac{(1-\gamma)}{\rho}} \right]^{\frac{1}{1-\gamma}}} \right)^{1-\gamma-\rho}. \end{aligned} \quad (25)$$

We guess

$$v_{j,t} = \frac{1}{\rho} (a_{j,t} \omega_{j,t})^\rho.$$

So the certainty equivalent can be written as

$$E_t \left[v_{j,t+1}^{\frac{(1-\gamma)}{\rho}} \right]^{\frac{1}{1-\gamma}} = \left(\frac{1}{\rho} \right)^{\frac{1}{\rho}} E_t \left[\pi (a_{i,t} \omega_{i,t})^{1-\gamma} + (1-\pi) (a_{s,t} \omega_{s,t})^{1-\gamma} \right]^{\frac{1}{1-\gamma}}$$

We consider the case of savers; the investors case is exactly the same. Substitute the guess in (25)

$$M_{s,t+1} = \beta \left(\frac{c_{s,t+1}}{c_{s,t}} \right)^{\rho-1} \left(\frac{a_{s,t} \omega_{s,t}}{E_t \left[\pi (a_{i,t+1} \omega_{i,t+1})^{1-\gamma} + (1-\pi) (a_{s,t+1} \omega_{s,t+1})^{1-\gamma} \right]^{\frac{1}{1-\gamma}}} \right)^{1-\gamma-\rho}. \quad (26)$$

Now we use $\frac{c_{s,t+1}}{c_{s,t}} = \frac{c_{s,t+1}}{\omega_{s,t+1}} \frac{\omega_{s,t+1}}{\omega_{s,t}} \frac{\omega_{s,t}}{c_{s,t}} = \frac{1-\tilde{\zeta}_{s,t+1}}{1-\tilde{\zeta}_{s,t}} R_{t+1}^{ss} \tilde{\zeta}_{s,t}$. (Where we have used the identity $\frac{\omega_{s,t+1}}{\omega_{s,t}} = R_{t+1}^{ss} \tilde{\zeta}_{s,t}$.) Then we can plug (26) in the first-order conditions

$$\left(\frac{\tilde{\zeta}_{s,t}}{1-\tilde{\zeta}_{s,t}} \right)^{\rho-1} E_t \left[\beta ((1-\tilde{\zeta}_{s,t+1}) R_{t+1}^{ss})^{\rho-1} \left(\frac{a_{s,t} \omega_{s,t}}{E_t \left[\pi (a_{i,t+1} \omega_{i,t+1})^{1-\gamma} + (1-\pi) (a_{s,t+1} \omega_{s,t+1})^{1-\gamma} \right]^{\frac{1}{1-\gamma}}} \right)^{1-\gamma-\rho} R_{t+1}^s \right] = 1$$

And the last step is to substitute

$$\begin{aligned}
 a_{s,t} &= (1 - \zeta_{s,t})^{\frac{1-\rho}{\rho}} \\
 a_{i,t} &= (1 - \zeta_{i,t})^{\frac{1-\rho}{\rho}} \\
 \frac{\omega_{i,t+1}}{\omega_{s,t}} &= \zeta_{s,t} R_{t+1}^{si} \\
 \frac{\omega_{s,t+1}}{\omega_{s,t}} &= \zeta_{s,t} R_{t+1}^{ss} \\
 \frac{\omega_{i,t+1}}{\omega_{i,t}} &= \zeta_{i,t} R_{t+1}^{ii} \\
 \frac{\omega_{s,t+1}}{\omega_{i,t}} &= \zeta_{i,t} R_{t+1}^{is} \\
 R_{t+1}^s &= \pi R_{t+1}^{si} + (1 - \pi) R_{t+1}^{ss} \\
 R_{t+1}^i &= \pi R_{t+1}^{ii} + (1 - \pi) R_{t+1}^{is}
 \end{aligned}$$

Together with the FOCs for investor, they define a system of equations in (ζ_s, ζ_i) . $\square$

Verification. We now need to verify that our guess actually verifies the value function. In the way we wrote the recursion, it is clear to see that any pair $\zeta_{i,t}, \zeta_{s,t}$ that satisfies solving the system of difference equations guarantee the functional form guessed for v_i and v_s . Notice that the envelope condition is already embedded in the recursion.

Uniqueness. The proof of uniqueness of the policy functions consists of checking that the recursion is in fact a contraction. That is, that it satisfies the Blackwell's sufficient conditions. A general treatment of this technical aspect can be found in [Epstein and Zin \(1989\)](#).

Proof of proposition 4 (Decomposition). The proof consists in substituting $R_{t+1}^{ss}, R_{t+1}^{si}, R_{t+1}^{is}, R_{t+1}^{ii}$ in R_{t+1}^i and R_{t+1}^s . We start with R_{t+1}^i

$$\begin{aligned}
 R_{t+1}^i &= \pi R_{t+1}^{ii} + (1 - \pi) R_{t+1}^{is}, \\
 &= \pi \left(\frac{r_{t+1} + q_{t+1}^i \lambda}{q_t^R} \right) + (1 - \pi) \left(\frac{r_{t+1} + q_{t+1} \lambda}{q_t^R} \right), \\
 &= \pi \left[\frac{r_{t+1} + q_{t+1}^i \lambda}{q_t^R} - \frac{r_{t+1} + q_{t+1} \lambda}{q_t^R} \right] + \frac{r_{t+1} + q_{t+1} \lambda}{q_t^R}, \\
 &= \pi \left[\frac{(q_{t+1}^i - q_{t+1}) \lambda}{q_t^R} \right] + \frac{r_{t+1} + q_{t+1} \lambda}{q_t^R}, \\
 &= \pi \left[\frac{(\phi_{t+1} q_{t+1} + (1 - \phi_{t+1}) q_{t+1}^R - q_{t+1}) \lambda}{q_t^R} \right] + \frac{r_{t+1} + q_{t+1} \lambda}{q_t^R}, \\
 &= \pi \lambda \left[\frac{(\phi_{t+1} - 1) q_{t+1} + (1 - \phi_{t+1}) q_{t+1}^R}{q_t^R} \right] + \frac{r_{t+1} + q_{t+1} \lambda}{q_t^R}, \\
 &= -\pi \lambda \frac{(q_{t+1} - q_{t+1}^R) (1 - \phi_{t+1})}{q_t^R} + \frac{q_t r_{t+1} + q_{t+1} \lambda}{q_t^R}, \\
 &= -\pi \frac{\lambda (q_{t+1} - q_{t+1}^R) (1 - \phi_{t+1})}{q_t^R} + \left(\frac{(1 - \theta) q_t}{1 - \theta q_t} \right) \frac{r_{t+1} + \lambda q_{t+1}}{q_t}, \\
 &= l_{t+1} + \eta_t R_t^{ss},
 \end{aligned}$$

which is equation (19). The exact same steps for $R_{t+1}^s = \pi R_{t+1}^{ss} + (1 - \pi) R_{t+1}^{si}$ deliver equation (20). $\square$

Numerical solution. We start by constructing a tensor grid for the state space (A, K, Φ) , and the objective is to project all endogenous variables onto it. Our reported results are for $D_A = 14, D_K = 20, D_\Phi = 14$. So the grid has $D_A \times D_K \times D_\Phi = 3920$ points. Results are robust to a lower degree of coarseness in the grid.

Next, we guess values for the endogenous values. We pick a random number generator for the ξ_s and ξ_i at every point of the state space (an educated guess would be β). We guess the first value of the aggregate endogenous state variable $(K')^{(0)} = \beta(r + \lambda)K$. We then start an iterative loop to solve for the policy functions, until convergence. The loop is as follows:

1. We start from the steady-state level of capital without constraints, and compute the transition probabilities.
2. Compute expectations and obtain an updated vector for ξ_i, ξ_s .
3. Compute the excess demand in the equity market, to check whether the constraint is binding.
4. If there is excess demand, we compute the q that clears the equity market in each state. Otherwise, $q = 1$.
5. Compute the policy functions c_i, c_s, n'_i, n'_s, L and prices with the values we obtained for q, q^R, q^i and w .
6. Verify these policy functions clear the goods market.
7. Compute $(K')^{(1)} = [\pi n'_i + (1 - \pi) n'_s] K$, and iterate until converge, i.e. $\|(K')^{(i)} - (K')^{(i-1)}\| \leq tol$, where tol is a small number.

Invariant distribution. We construct the exogenous Markov chain in the following way. First, we discretize the state space of exogenous productivity, which follows an AR(1) (in logs). For this, we proceed in line with Tauchen (1985, 1991). This give us a matrix P^A , in which each element is $P_{i,j}^A = \Pr \{A_{t+1} = a_j | A_t = a_i\}$.

We then compute the joint Markov chain for liquidity and productivity. We denote this joint transition probability matrix as $P^{A,\Phi}$, where an element of the matrix represents $P_{i,j}^{A,\Phi} = \Pr \{\phi_{t+1} = \phi_j, A_{t+1} = a_j | \phi_t = \phi_i, A_t = a_i\}$ (abusing notation, i is a pair) and is given by

$$P_{i,j}^{A,\Phi} = P_{i,j}^A \left(\varphi P_{i,j}^{A(\phi')} + (1 - \varphi) U_{i,j} \right),$$

where $P_{i,j}^{A(\phi')}$ is an auxiliary matrix we construct to capture the correlation between productivity and liquidity, given by φ . This mapping is simply the order of elements in the liquidity vector Φ , and $U_{i,j}$ is a uniform probability matrix. $P^{A,\Phi}$ has all the desired properties of a transition probability matrix.

The solution of the model gives us a transition matrix $P^{K',A,\Phi} = \mathbb{P}$, where an element is

$$P_{i,j}^{K',A,\Phi} = \Pr \{K_{t+1} = K_j; \phi_{t+1} = \phi_j, A_{t+1} = a_j | K_t = K_i, \phi_t = \phi_i, A_t = a_i\}$$

Then, the invariant distribution, μ , simply solves the following eigenvalue problem

$$\mu' (I - \mathbb{P}) = 0$$

where I is the identity matrix and the vector¹² $\mu = \lim_{t \rightarrow \infty} \mu_t = \mathbb{P}(K_\infty, A_\infty, \phi_\infty)$, with $\mu'_{t+1} = \mu'_t \mathbb{P}$. This vector has dimension $(1 \times (N_K \times N_A \times N_\phi))$. The problem is guaranteed to have a solution because $\mathbb{P}$ is a Markov chain (positive semi-definite and rows sum to 1), so we select the eigenvector associated with the (unique) unitary eigenvalue.

¹²We normalize the vector to ensure that the sum of its elements is equal to one.

Then, we can compute unconditional theoretical moments of endogenous variables using this unconditional measure, following theorem 2.2.3 in [Ljungqvist and Sargent \(2012\)](#). That is, for any random variable $z(s)$, we compute the unconditional mean as $\mathbb{E}_\mu [z(s)] = \sum_i^N \mu_i z_i$ (where μ is normalized to be a probability measure.)

We compute the conditionally constrained distribution as follows. Let $\mathcal{C} = \{s \in S : q(s) > 1\}$ be the set in which constraints bind (i.e., the spot price of capital is greater than one). Then, we define the following random variable (an indicator function):

$$1_{\mathcal{C}}(s) = \begin{cases} 1 & \text{if } s \in \mathcal{C} \\ 0 & \text{if } s \notin \mathcal{C} \end{cases}$$

Then

$$\tilde{\mu} = \mathbb{P}(K_\infty, A_\infty, \phi_\infty | \mathcal{C}) = \frac{\mathbb{E}_\mu [1_{\mathcal{C}}(s)]}{\sum_{i:s \in \mathcal{C}} \mu_i}$$

We can compute the conditional expectation numerically, by constructing this indicator function across the state space. Then we can compute, for example, the conditionally constrained mean for a given random variable z as $\mathbb{E}_{\tilde{\mu}} [z(s)] = \sum_i^N \tilde{\mu}_i z_i$. Figure (11) displays the results for μ and $\tilde{\mu}$ (each bar represents the size of the measure for each point in the K state). The conditionally unconstrained distribution can be computed in a similar way, by simply defining the complement of $\mathcal{C}$. In Table 4 we report statistics using these three measures.

Impulse-response functions. We construct the impulse response functions (IRF) using a standard simulation method. In Figure (12), we report the IRFs to a liquidity shock. We start the economy in the unconditional means. We then simulate multiple paths of the exogenous Markov chain to obtain a sequences of $\{A_t^{(j)}, \phi_t^{(j)}\}_{t=0}^T$, in each simulation $j = 1, \dots, J$. We simulate these paths with $\phi_0 = 0.05$ (i.e., in period 0 there is a liquidity shock that moves liquidity from 0.275 to 0.5, as mentioned in the text). In each simulation, we use the model solution to obtain the responses (i.e., the paths) of endogenous variables in each simulation. We then average paths across simulations. That is, for a given variable z , we have J simulated paths $\{z_0^{(j)}, z_1^{(j)}, \dots, z_T^{(j)}\}_{j=1}^J$, so we take the mean $\frac{1}{J} \sum_{j=1}^J z^j = z^{irf}$, and we obtain the sequence z^{irf} .